Notes on Affine Grassmannians

Jiangfan Y.

Contents

1 First Glimpse at Affine Grassmannians

1.1 G-torsors on a disc

Recall that a 'canonical' Grassmannian $G(l,n)$ over arbitrary scheme S is a presheaf $\mathrm{Sch}/S \to \mathrm{Set}$, mapping

$$S' \mapsto \left\{ \text{isom. classes of } q : \mathcal{O}_{S'}^{\oplus n} \twoheadrightarrow \mathcal{Q} | \mathcal{Q} \text{ locally free of rank } n-l \right\}$$

Moreover, it is representable by a projective scheme (see for example [Sta, Tag 089R]) . In other language, it is a homogeneous space $\mathrm{GL}_n/\mathrm{P}_{n-l,l}$. When the base scheme is given by $\mathrm{Spec}\,\mathbf{k}$, $\mathbf{k}$ field, then this Grassmannian parametrizes flags of type $(l, n-l)$ in k^n. Take $\mathrm{G}(n)$ to be the disjoint union $\coprod_{0\leqslant l\leqslant n} \mathrm{G}(l,n)$.
The affine Grassmannians (for GL_n) can be constructed in similar manner to that of $\mathrm{G}(n)$, but in the infinite level. We start with a field $\mathbf{k}$, write $F = \mathbf{k}((t))$ the field of Laurent series and $\mathcal{O} = \mathbf{k}[[t]]$ ring of formal series, as well as ring of integers of F. Recall a lattice in F^n is a finitely generated $\mathcal{O}$-submodule Λ of F^n so that $\Lambda \otimes_{\mathcal{O}} F \simeq F^n$. For R any $\mathbf{k}$-algebra, substitute $F, \mathcal{O}$ with $R((t)), R[[t]]$ one can define lattices for R.

Definition 1.1. The GL_n affine Grassmannian over $\mathbf{k}$ is a presheaf $\mathrm{Alg}_{\mathbf{k}}^{\mathrm{op}} \to \mathrm{Set}$ assigns

$$R \mapsto \{R[[t]] - \text{Lattices in } R((t))^n\}.$$

We denote it by $\mathrm{AffGr}_{\mathrm{GL}_n}$, abbreviation: Gr_n.

The presheaf defined above however is not representable by a scheme. Instead, it is a space (with respect to fpqc topology). Moreover, it is in fact an (strict)ind-projective scheme. To be precise,

Definition 1.2. An **ind-scheme** is an ind-object in the category of schemes, by ind-object we mean a filtered colimit. By strictness we mean if an ind-scheme $X := \mathrm{colim}_{i\in I} X_i$, then for any $i \leqslant j$, the corresponding map $X_i \to X_j$ is represented by a closed embedding. An ind-$\mathbf{P}$ scheme is an ind-scheme $X := \mathrm{colim}_{i\in I} X_i$ with each X_i being scheme of property $\mathbf{P}$.

Remark. An ind-scheme is automatically an fpqc space (i.e satisfies sheaf conditions) because one takes the filtered colimit.

Return to the case of GL_n affine Grassmannian, the filtered system is given by

$$\mathrm{Gr}_n^{[a,b]} : R \mapsto \left\{\Lambda | t^b R[[t]] \subseteq \Lambda \subseteq t^a R[[t]]\right\}$$

where $a \leq b$ integers. Or alternatively, we can take more coarse ones

$$\mathrm{Gr}_n^{(N)} : R \mapsto \left\{\Lambda | t^{-N} R[[t]] \subseteq \Lambda \subseteq t^N R[[t]]\right\}$$

It remains to prove

Theorem 1.3. $\mathrm{Gr}_n^{(N)}$ *is represented by a closed subscheme of* $\mathrm{G}(2nN)$ *(the usual Grassmannian).*

Proof. For $\Lambda \in \mathrm{Gr}_n^{(N)}(R)$, consider the well defined quotient $Q := t^{-N}R[[t]]^n/\Lambda$, it is a projective R module. In fact, Q fits into a short exact sequence

$$0 \longrightarrow Q \longrightarrow R((t))^n/\Lambda \longrightarrow R((t))^n/t^{-N}R[[t]]^n \longrightarrow 0$$

Note that $R((t))^n/\Lambda \simeq \Lambda \otimes_{R[[t]]} R((t))/\Lambda \simeq \Lambda \otimes_{R[[t]]} \left(\bigoplus_{k\geq 0} t^{-k-1}R[[t]]/t^{-k}R[[t]]\right)/\Lambda \simeq \bigoplus_{k\geq 0} t^{-k-1}\Lambda/t^{-k}\Lambda$. Thus the last two terms of the short exact sequence are projective (as R-module), we can deduce Q is also projective.

Define $\mathrm{Gr}_n^{(N),f}$ to be the presheaf so that

$$\mathrm{Gr}_n^{(N),f}(R) = \left\{R[[t]]\text{-quotient modules of } t^{-N}R[[t]]^n/t^N R[[t]]^n \text{ that are projective as } R\text{-modules}\right\}$$

It is clear that $\mathrm{Gr}_n^{(N),f}$ is representable by a closed subscheme of $\mathrm{Gr}(2nN)$ since $t^{-N}R[[t]]^n/t^N R[[t]]^n \simeq R^{2nN}$ as R-module.

Now $\Lambda \mapsto Q$ gives a morphism of presheaves $\mathrm{Gr}_n^{(N)} \to \mathrm{Gr}_n^{(N),f}$. It remains to show that this gives an isomorphism.

For a given $Q \in \mathrm{Gr}_n^{(N),f}(R)$, associate $\Lambda_Q := \mathrm{Ker}(t^{-N}R[[t]]^n \to t^{-N}R[[t]]^n/t^N R[[t]]^n \to Q)$, it reduces to show that Λ_Q is finitely generated projective $R[[t]]$-module. In fact, this comes from reduction of Λ_Q to an $R[t]$-module. We can further assume that R is Noetherian $\mathbf{k}$-algebra, as $\mathrm{Gr}_n^{(N),f}$ is locally of finite presentation, hence commute with directed colimit. Thus we can restrict to Noetherian piece R_i of R and glue together by taking colim_i. With Noetherianity, we can prove $R[[t]]$ (t-adic completion) is flat as $R[t]$-module by using Baer criterion (see for example [Sta, Tag 00MB]), take

$$\Lambda_Q^0 := \mathrm{Ker}(t^{-N}R[t]^n \to t^{-N}R[t]^n/t^N R[t]^n \to Q)$$

here $t^{-N}R[t]^n/t^N R[t]^n \simeq t^{-N}R[[t]]^n/t^N R[[t]]^n$ by flatness. and we know $\Lambda_Q^0 \otimes_{R[t]} R[[t]] \simeq \Lambda_Q$. Hence Λ_Q is projective if Λ_Q^0 is flat $R[t]$-module.

To prove Λ_Q^0 is flat $R[t]$ module, need to show for any ideal $\mathfrak{p} \in R[t]$, $(\Lambda_Q^0)_\mathfrak{p}$ is $R[t]_\mathfrak{p}$ flat. We first see for $\forall \mathfrak{q} \in \mathrm{Spec}\, R$, $\Lambda_Q^0 \otimes_{R[t]} \kappa_\mathfrak{q}[t]$ is flat as $\kappa_\mathfrak{q}[t]$ module, where $\kappa_\mathfrak{q}$ is the corresponding residue field. In fact, $\kappa_\mathfrak{q}[t]$ is Dedekind domain, and it is clear that $\Lambda_Q^0 \otimes_{R[t]} \kappa_\mathbf{q}[t]$ is torsion free.

With above justification, take arbitrary $\mathfrak{p} \in \mathrm{Spec}\, R[t]$, $\mathfrak{q}$ the inverse image of $\mathfrak{p}$ under $R \to R[t]$, we have

$$(\Lambda_Q^0 \otimes_{R[t]} \kappa_\mathfrak{q}[t])_\mathfrak{p} \simeq (\Lambda_Q^0)_\mathfrak{p}/\mathfrak{q}(\Lambda_Q^0)_\mathfrak{p} \text{ is } (\kappa_\mathfrak{q}[t])_\mathfrak{p} \simeq R_\mathfrak{q}[t]_\mathfrak{p}/\mathfrak{q}R_\mathfrak{q}[t]_\mathfrak{p} \simeq (R[t]/\mathfrak{q}R[t])_\mathfrak{p} \text{ flat module}$$

With obvious finiteness, it is thus projective $(\kappa_\mathbf{q}[t])_\mathfrak{q}$ module. As projective modules over local ring are free, we can apply [Sta, Tag 00MH] to deduce that $(\Lambda_Q^0)_\mathfrak{p}$ is a free $R[t]_\mathfrak{p}$ module, hence flat. □

Let $D_\mathbf{k}$ and $D_\mathbf{k}^*$ denote the unit disc and punctured disc respectively, for $R \in \mathbf{k}\,\mathrm{Alg}$, denote also D_R, D_R^* base changed discs. Then to give the data of $R[[t]]$-lattice in $R((t))^n$ is amount to give a vector bundle $\mathcal{E}$ (i.e GL_n-torsor) with a trivialization β on D_R^* (on the level of module, a finitely generated $R[[t]]$ module Λ such that $\beta : \Lambda \otimes_{R[[t]]} R((t)) \xrightarrow{\simeq} R((t))^n$).

Now let G be any smooth affine group scheme over $\mathbf{k}$, we can generalize the definition of affine Grassmannian:

Definition 1.4. The affine Grassmannian Gr_G is defined as the functor (presheaf) that maps

$$R \in \mathbf{k}\,\mathrm{Alg} \mapsto \left\{(\mathcal{E}, \beta) \mid \mathcal{E} \text{ is a } G\text{-torosr on } D_R, \beta \text{ a trivialization on } D_R^*.\right\}$$

Theorem 1.5. *The presheaf* Gr_G *is representable by ind-finite type scheme over* $\mathbf{k}$. *In particular, when* G *is reductive,* Gr_G *is an ind-projective scheme.*

Proof. We want an embedding of presheaves $\mathrm{Gr}_G \to \mathrm{Gr}_n$, this is realized by finding G a faithful representation.

Lemma 1.6. *For smooth affine* $\mathbf{k}$ *group scheme* G, *there is a* $\mathbf{k}[[t]]$ *faithful representation* $G \to \mathrm{GL}_n$. *Moreover, if* $H \hookrightarrow G$ *is a closed embedding of smooth group scheme into reductive group* G, *then the quotient* G/H *is quasi-affine. When further* H *is reductive,* G/H *is affine.*

Fix $G \to \mathrm{GL}_n$ as above Lemma, We have a map $\mathrm{Gr}_G \to \mathrm{Gr}_n$, $(\mathcal{E}_0, \beta_0) \mapsto (\mathcal{E}_0 \times^G \mathrm{GL}_n, \beta)$, where β denotes the lift of trivialization along $\mathcal{E}_0 \times^G \mathrm{GL}_n$. Need to show it is represented by a closed embedding. Take a section $S = \operatorname{Spec} R \to \mathrm{Gr}_n$, represented by $(\mathcal{E}, \beta)$, we consider the fiber product $S \times_{\mathrm{Gr}_n} \mathrm{Gr}_G$, it is a presheaf that assigns $R' \to R$ to the set of reductions of $(\mathcal{E}, \beta)_{R'}$.
Let $\pi : \mathcal{E} \to D_R$ be the projection, then β is a section of π over D_R^*. Recall the (fpqc) quotient stack $[\mathcal{E}/G]$ assigns S' to groupoid with objects $S' \xleftarrow{p} E \xrightarrow{\phi} \mathcal{E}$, where p is a $G_{S'}$-torsor and ϕ is G-equivariant. Thus to give a reduction of $\mathcal{E}$ is to give a section of $\bar{\pi} : [\mathcal{E}/G] \to D_R$. Also we can regard trivialization β as a section $D_R^* \to \mathcal{E}$, which naturally induces $\bar{\beta} : D_R^* \to [\mathcal{E}/G]$, hence a reduction of $(\mathcal{E}, \beta)$ requires a section β' coincides with $\bar{\beta}$ on punctured disc. This can be seen from the following diagrams:

$$\begin{tikzcd}
\mathcal{E} \arrow[r] \arrow[dr, "\pi"] & {[\mathcal{E}/G]} \arrow[d, "\bar{\pi}"] \\
D_R^* \arrow[u, "\beta"] \arrow[r, hook, "\iota"'] & D_R
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{E} \arrow[r] & {[\mathcal{E}/G]} \\
D_R^* \arrow[u, "\beta"] \arrow[ur, "\bar{\beta}"] \arrow[r, hook, "\iota"'] & D_R \arrow[u, dashed, "\beta'"']
\end{tikzcd}$$

Since fpqc descents of (quasi-)affine are effective, we can find $[\mathcal{E}/G]$ a affine scheme W of finite presentation over D_R so that $j : [\mathcal{E}/G] \hookrightarrow W$ is representable by open immersion and is quasi-compact. Embed W in $\mathbb{A}^N_{D_R}$ for $N \gg 0$, extend $\bar{\beta}$ to $j \circ \bar{\beta}$, consider the presheaf

$$R'/R \mapsto \left\{ \text{sections } \beta'' \text{ over } D_{R'} \text{ so that } \beta''|_{D^*_{R'}} = (j \circ \bar{\beta})|_{D^*_{R'}} \right\}$$

is represented by a closed subscheme of $\operatorname{Spec} R$, as we can write $j \circ \bar{\beta}$ in affine coordinate $\big(j \circ \bar{\beta}_1(t), ..., j \circ \bar{\beta}_N(t)\big)$, with $j \circ \bar{\beta}_i(t) = \sum_k r_{ik} t^k \in R((t))$, and we take the closed subscheme of R corresponding to the ideal generated by $r_{ik}, k \le 0$. Denote this subscheme by $\operatorname{Spec} R_0$, in other words, R_0 is the locus over which $j \circ \bar{\beta}$ extends to a section over full disc D_{R_0}. Finally, pulling back along the evaluation map $t \mapsto 0$ with respect to j, we get a open subscheme of Spec R_0 that represents Spec $R \times_{\mathrm{Gr}_n} \mathrm{Gr}_G$. To be more precisely, we have the following diagram:

$$\begin{tikzcd}
 & D^*_{R_0} \arrow[r, hook] \arrow[d, hook] & D^*_R \arrow[r, "\bar{\beta}"] \arrow[d, hook] & {[\mathcal{E}/G]} \arrow[d, hook, "j"] \\
\mathrm{Spec}\, R_0 \arrow[r, "ev"] \arrow[drr, "\beta''_0"'] & D_{R_0} \arrow[r, hook] \arrow[rr, bend right, "\beta''"'] & D_R & W \arrow[l] \\
\beta''^{-1}_0([\mathcal{E}/G] \times_{D_R} R_0) \arrow[r] \arrow[u, hook] & {[\mathcal{E}/G] \times_{D_R} R_0} \arrow[r, hook] & W \times_{D_R} R_0 \arrow[ur] &
\end{tikzcd}$$

where β'' is the section on the locus D_{R_0} coincides with $j \circ \bar{\beta}$ on $D^*_{R_0}$ and β''_0 is the section of R_0 induced by β'' along the evaluation map ev. □

Remark. Throughout the above discussions, we take G to be a smooth affine $\mathbf{k}$ group. Observe that the presheaf of affine Grassmannian can be defined for smooth affine $\mathbf{k}[[t]]$ group schemes. In fact, the statements of Lemma 1.6 and thus Theorem 1.5 still hold in this case, see [Zhu16] for a discussion.

Remark. Ind-projectivity is an important property of affine Grassmannians, which is inherited by analogs for variations of affine Grassmannians. For example, the Witt vector affine Grassmannians are ind-perfectly projective, proved in [BS17], and the B^+_{dR}-affine Grassmannians are proper v-spaces as in [SW20]. We will see them later.

1.2 Loop groups

Another formulation of affine Grassmannians is given by loop groups. Namely we want $\mathrm{Gr}_G(\mathbf{k}) := G(\mathbf{k}((t)))/G(\mathbf{k}[[t]])$. For simplicity, we consider any X scheme over $\mathbf{k}$, we can define its loop space (resp. arc space) to be the presheaf LX (resp. L^+X) such that

$$LX(R) = X(R((t))); \qquad L^+X(R) = X(R[[t]]), R \in \mathbf{k}\,\mathrm{Alg}$$

When X is affine scheme of finite type, the arc space L^+X is an (infinite) affine scheme, viewed as projective limit of jet scheme L^nX. The construction of arc and jets is formal, we assume $X = \mathrm{Spec}\,\mathbf{k}\,[x_1, ..., x_N]\,/(f_1, ..., f_m)$. Now formally extend each coordinate x_i to $x_i(t) := \sum_{j=1}^{\infty} x_i^{(j)} t^{j-1}$ and it transforms each $f(x_1, ..., x_N) = 0$ to a system equation $f(x_1(t), ..., x_N(t)) = 0$. Then for $n \in \mathbb{N}$, $L^nX \subseteq \mathrm{Spec}\,\mathbf{k}[x_i^{(j)}; i = 1, ..., N, j \leqslant n]$ is defined by $f_k(x_1(t), ..., x_N(t)) = 0 \bmod t^n; k = 1, ..., m$. Clearly those L^nX are affine and $L^+X \subseteq \prod_{j=1}^{\infty} \mathbb{A}^N$ is given by $f_k(x_i(t)) = 0$. We see thus L^+X is affine.

The loop space LX however is not affine scheme, under our assumption that X finite type affine scheme over $\mathbf{k}$, it is ind-affine scheme of finite type. For example,

$$L\mathbb{A}^N = \varinjlim_{j} \mathrm{Spec}\,\mathbf{k}\left[x_i^{(-j)}, x_i^{(-j+1)}, \ldots\right]_{i=1,\ldots,N}$$

here $x_i^{(-j)}$ denotes the coefficient in the Laurent series. For full generality, see for example [KV04], [PR08]. Throughout, we need only the loop (arc) space for $X = G$. In that case, L^nG, L^+G are affine group schemes, and we have

$$\mathrm{Lie}(L^+G) = \mathfrak{g}[[t]] = \varprojlim_n \left(\mathrm{Lie}(L^nG) \simeq \mathfrak{g}[t]/(t^{n+1})\right)$$

where $\mathfrak{g} = \mathrm{Lie}(G)$ is the Lie algebra of G and the Lie bracket on $\mathfrak{g}[[t]] = \mathfrak{g} \otimes_{\mathbf{k}} \mathbf{k}[[t]]$ is given by $[x \otimes t^m, y \otimes t^n] = [x, y] \otimes t^{m+n}$.

The terminology 'loop group' also makes sense in a view of homotopy, see [PR08][Theorem 0.1] demonstrates when $\mathbf{k}$ algebraically closed $\pi_0(LG) \simeq \pi_1(G)_I$, where $I = \mathrm{Gal}(\bar{\mathbf{k}((\mathbf{t}))}/\mathbf{k}((t)))$ is the inertia group, and $\pi_1(G)_I$ is the co-invariant part.

Turning back to affine Grassmannians, we have the formalism by loop group:

Proposition 1.7. *The affine Grassmannian* Gr_G *associated with G is isomorphic to the quotient stack* $[LG/L^+G]$ *(as fpqc sheaf).*

Remark. Note that when G is smooth affine group scheme, G-torsors are étale locally trivialisable. If G is reductive, then G-torsors are further Zariski locally trivialisable.

Proof. For any $R \in \mathbf{k}\,\mathrm{Alg}$, by remark above, there is $R' \to R$ étale so that G-torsors are trivialisable over $D_{R'}$. Thus we can reduce to the case of trivialisable torsors.the fiber $LG(R)$ can be identified with the setoid of triples

$$\left\langle (\mathcal{E}, \beta, \epsilon) \mid (\mathcal{E}, \beta) \in \mathrm{Gr}_G(R); \epsilon : \mathcal{E} \xrightarrow{\simeq} G \times D_R \text{ trivialization over disc } D_R \right\rangle$$

In fact, the isomorphism is established by $LG(R) = G(R((t))) \ni M \mapsto (G \times D_R, M, \mathrm{id})$ and on the converse side we have map $(\mathcal{E}, \beta, \epsilon) \mapsto M := \beta^{-1} \circ \epsilon \in \mathrm{Aut}(\mathcal{E} \mid_{D_R^*}) = G(R((t)))$ □

Another variation of loop group is the opposite (or negative) loop group L^-G representing the functor $L^-G(R) = G(R[t^{-1}])$. Similar to previous description, it is an ind-affine group scheme. When G is reductive, recall there is a Weil dictionary

$$\mathrm{Bun}_G(\mathbb{P}^1)(\mathbf{k}) = G(\mathbf{k}[t^{-1}])\backslash G(\mathbf{k}((t)))/G(\mathbf{k}[[t]])$$

Simply from the fact that G torsors on $\mathbb{P}^1$ can be trivialized on $\mathbb{P}^1\backslash\infty$. Such isomorphism motivates the isomorphism of stacks

Proposition 1.8. *When G is connected reductive group, there is an isomorphism* $[L^-G\backslash \mathrm{Gr}_G] \simeq \mathrm{Bun}_G(\mathbb{P}^1)$

1.3 Cartan decomposition

Use the notation $F = \mathbf{k}((t)), O = \mathbf{k}[[t]]$ as in section 1.1, assume also $\mathbf{k}$ is algebraically closed. Fix $T \subseteq B \subseteq G$ maximal torus and Borel of G, recall the classical Cartan decomposition

$$G(F) = \coprod_{\mu \in \mathbb{X}_*(T)^+} G(O) \cdot t^\mu \cdot G(O) \quad \text{or equivalently} \quad LG(\mathbf{k}) = \coprod_{\mu \in \mathbb{X}_*(T)^+} L^+G(\mathbf{k}) \cdot t^\mu \cdot L^+G(\mathbf{k})$$

where $\mathbb{X}_*(T)$ is the coweight lattice $\mathrm{Hom}(\mathbb{G}_m, T)$, and $\mathbb{X}_*(T)^+$ is the dominant points for arbitrary R. For $\mu \in \mathbb{X}_*(T)$, t^μ is its image in $T(F)$ under $F^\times \longrightarrow T(F)$.

Roughly speaking, take the right quotient by $G(O)$, we get a 'cell decomposition' of $\mathrm{Gr}(\mathbf{k})$ by $\mathbb{X}_*(T)^+$, each cell labeled by μ is called a Schubert cell, as an analog to flag varieties.

Before proceeding to discuss geometric property of Schubert varieties, we give a geometric proof of Cartan decomposition, following the method in [AHH19].

Theorem 1.9. *Let R be a complete DVR, with fraction field F and an algebraically closed residue field $\mathbf{k}$, fix some uniformizer ξ. Then for any reductive group G over O, we have $G(F) = \coprod_\mu G(R) \cdot \xi^\mu \cdot G(R)$*

Proof. Consider the affine scheme $Y = \mathrm{Spec}\, R[s,t]/(st - \xi)$ with $\mathbb{G}_m$ action on parameters s, t of weight $1, -1$ respectively. Denote by 0 the maximal ideal $\mathfrak{m} := (s,t)$. We know $Y \backslash \{0\}$ is glued by two pieces $Y_s = \mathrm{Spec}\, R[s]_s \simeq \mathrm{Spec}\, R \times \mathbb{G}_m$ and $Y_t = \mathrm{Spec}\, R[t]_t \simeq \mathrm{Spec}\, R \times \mathbb{G}_m$ along $Y_{st} \simeq \mathrm{Spec}\, F \times \mathbb{G}_m$.

First, we take trivial G torsors on Y_s and Y_t respectively, any $g \in G(F)$ gives an isomorphism of those two trivial torsors restricting on Y_{st}, thus a gluing to a G torsor, say $\mathcal{P}_g$, on $Y \backslash \{0\}$. Moreover, such an isomorphism by g is $\mathbb{G}_m$-equivariant by obvious left $\mathbb{G}_m$ action on $Y_s \simeq \mathrm{Spec}\, R \times \mathbb{G}_m$ (resp. Y_t), hence $\mathcal{P}_g \to Y \backslash \{0\}$ is endowed with a left $\mathbb{G}_m$ action compatible with $\mathbb{G}_m$ action on the base $Y \backslash \{0\}$. Note also there is a right G action on $\mathcal{P}_g$ by simultaneous right multiplication on trivial G-torsors over charts Y_s, Y_t, the left $\mathbb{G}_m$ action commutes with G action on the right.

The torsor $\mathcal{P}_g$ constructed above can be extended uniquely to whole Y, say $\tilde{\mathcal{P}}_g$. To give $\mathcal{P}_g$ is the equivalent to give the data of a GL_n torsor (or a vector bundle) $\mathcal{E} \to Y \backslash \{0\}$ and a section $Y \backslash \{0\} \to \mathcal{E}/G$. Since Y is regular and the point 0 is of codimension 2, then by Serre's criterion, $\mathcal{E}$ is uniquely extended on Y to $\tilde{\mathcal{E}}$, where $\tilde{\mathcal{E}}$ is simply taken as $(Y \backslash \{0\} \hookrightarrow Y)_* \mathcal{E}$. We get naturally a section $Y \backslash \{0\} \to \tilde{\mathcal{E}}/G$, it can be extended to a section $Y \to \tilde{\mathcal{E}}/G$ since $\tilde{\mathcal{E}}/G$ is affine and $\Gamma(Y, O_Y) = \Gamma(Y \backslash \{0\}, O_Y)$. Moreover, the left $\mathbb{G}_m$ action and the right G action are inherited, compatible with $\mathbb{G}_m$ action on the whole base Y.

On the other hand, for any one parameter subgroup $\lambda : \mathbb{G}_{m,R} \to G$ we can consider a trivial G torsor on Y with left $\mathbb{G}_m$ action by λ, denote by $\mathcal{P}_\lambda$. The restriction of $\tilde{\mathcal{P}}_g$ to $0 \simeq \mathrm{Spec}\, \mathbf{k}$ is a trivial $G_\mathbf{k}$ torsor. The $\mathbb{G}_m$ action, under restriction, gives a morphism $\lambda_0 : \mathbb{G}_{m,\mathbf{k}} \to \mathrm{Aut}(\tilde{\mathcal{P}}_\} \mid_{\{0\}}) \simeq G_\mathbf{k}$. Since R is complete, λ_0 can be extended to $\lambda : \mathbb{G}_{m,R} \to G$. For such λ obtained, we claim there is a $\mathbb{G}_m$ equivariant isomorphism of G torsors between $\tilde{\mathcal{P}}_\}$ and $\mathcal{P}_\lambda$.

Proof of Claim. Consider the sheaf $\mathcal{I} := \underline{\mathrm{Isom}}(\mathcal{P}_\lambda, \mathcal{P}_g)$, the claim is equivalent to say there is a $\mathbb{G}_m$ equivariant global section $Y \to \mathcal{I}$. Let $Y^{[n]} \subseteq Y$ denote the n-th thickening of the origin, i.e the subscheme of Y defined by

$\mathfrak{m}^{n+1}$. On the level 0, we have an isomorphism $\alpha^{[0]}: \mathcal{P}_\lambda\mid_{\{0\}} \xrightarrow{\simeq} \tilde{\mathcal{P}}_\}\mid_{\{0\}}$. As $\underline{\mathrm{Isom}}(\mathcal{P}_\lambda, \mathcal{P}_g)$ is Zariski locally isomorphic to G and G is smooth, we know $\alpha^{[0]}$ can be extended to any finite order thickening $\alpha^{[n]}: \mathcal{P}_\lambda\mid_{Y^{[n]}} \xrightarrow{\simeq} \tilde{\mathcal{P}}_\}\mid_{Y^{[n]}}$, eventually we get a family of liftings.

Also, since $\mathbb{G}_m$ is linearly reductive, the above family of liftings is $\mathbb{G}_m$ equivariant on each level. (?).

Write $A := \Gamma(Y, O_Y) = R[s,t]/(st-\xi)$, then each $Y^{[n]}$ corresponds to $A^{[n]} := A/\mathfrak{m}^{n+1}$. To give a global section is to give a map $\Gamma(\mathcal{I}, O_{\mathcal{I}}) \to A$. And the $\mathbb{G}_m$ action on $s,t \in A$ by weight ± 1 induces a $\mathbb{Z}$ grading on A and respectively on $A^{[n]}$:

$$A = \oplus_d A_d,\ \text{where}\ A_d = \begin{cases} R\left\langle s^d \right\rangle & d \geq 0 \\ R\left\langle t^d \right\rangle & d < 0 \end{cases} \text{ and } A^{[n]} = \oplus_d A_d^{[n]},\ \text{where}\ A_d^{[n]} = \begin{cases} R/\left(\pi^{\lfloor \frac{n-d}{2} \rfloor + 1}\right)\left\langle s^d \right\rangle & n \geq d \geq 0 \\ R/\left(\pi^{\lfloor \frac{n+d}{2} \rfloor + 1}\right)\left\langle t^d \right\rangle & n \geq -d > 0 \\ 0 & \text{other} \end{cases}$$

To have a global section that is $\mathbb{G}_m$ equivariant requires $\Gamma(\mathcal{I}, O_{\mathcal{I}}) \to A$ to be graded, with respect to $\mathbb{Z}$-grading on $\Gamma(\mathcal{I}, O_{\mathcal{I}})$ given also by $\mathbb{G}_m$ action. Now since R is complete, we have $A_d = \varprojlim_n A_d^{[n]}$. The family $\alpha^{[n]}$ gives each $d \in \mathbb{Z}$ compatible maps $\Gamma(\mathcal{I}, O_{\mathcal{I}})_d \to A_d^{[n]}$, hence a map $\Gamma(\mathcal{I}, O_{\mathcal{I}})_d \to A_d$. The claim is proved. □

To finish the proof of Cartan decomposition, observe that $g_1, g_2 \in G(K)$ determines a same G torsor on $Y\backslash\{0\}$ if and only if there exists $h_1, h_2 \in G(R)$ such that $g_1 = h_1 g_2 h_2$. □

Remark. The condition that $\mathbf{k}$ being algebraically closed is subtle throughout the proof. In fact, such condition is shown to be redundant in [AHH19], so it is harmless to write Cartan decomposition for $R = \mathbf{k}[[t]]$ for any field $\mathbf{k}$.

Formally, fix G a reductive group with maximal torus T, for $\mu \in \mathbb{X}_*(T)^+$ we can define sub-presheaves $\mathrm{Gr}_\mu \subseteq \mathrm{Gr}_G$ to be:

$$\forall R \in \mathbf{k}\,\mathrm{Alg}, \mathrm{Spec}\, R \to \mathrm{Gr}_G \text{ factors through } \mathrm{Gr}_\mu \text{ if } \forall x \in \mathrm{Spec}\, R, \text{ corresponding } \kappa(x) \text{ valued point lies in } L^+G(\kappa(x)) \cdot t^\mu$$

and we can similarly define $\mathrm{Gr}_{\leqslant\mu}$ by replacing $L^+G(\kappa(x)) \cdot t^\mu$ above with the union

$$\coprod_{\mathbb{X}_*(T)^* \ni \mu' \leqslant \mu} L^+G(\kappa(x)) \cdot t^{\mu'}$$

In fact, the presheaves defined above are finite dimensional varieties, shown in [Zhu16].

We first introduce the notion of relative position of a modification. Let $\mathcal{E}_1, \mathcal{E}_2$ be two G-torsors over $D_{\mathbf{k}}$, trivialized over $D_{\mathbf{k}}^*$, denote also $\mathcal{E}^0$ the trivial G torsor (for convenience we take $\mathbf{k}$ to be algebraically closed). For $\beta: \mathcal{E}_1\mid_{D_{\mathbf{k}}^*} \xrightarrow{\simeq} \mathcal{E}_2\mid_{D_{\mathbf{k}}^*}$ and $\phi_i \in \underline{\mathrm{Isom}}(\mathcal{E}_i, \mathcal{E}^0), i = 1,2$, we know $\phi_2 \circ \beta \circ \phi_1^{-1} \in \mathrm{Aut}\left(\mathcal{E}^0\mid_{D_{\mathbf{k}}^*}\right) = G(\mathbf{k}((t)))$, varying ϕ_i is equivalent to automorphisms on $\mathcal{E}^0$, hence left and right multiplication by $G(\mathbf{k}[[t]]) = \mathrm{Aut}_{D_{\mathbf{k}}}(\mathcal{E}^0)$. According to Theorem 1.9 of Cartan decomposition and the remark follows, we know β corresponds to an element

in $G(O)\backslash G(F)/G(O) \simeq \mathbb{X}_*(T)^+$, denote this element by $\mathrm{Inv}_{\mathbf{k}}(\beta)$.
The above construction has rigidity on field extension, as explained in [Zhu16] Remark 2.1.3, if a (not necessarily algebraically closed) field K contains $\mathbf{k}$, then pass to algebraic closure $\bar{K}$ we can define

$$\mathrm{Inv}_{\bar{K}}(\beta_{\bar{K}}) \in G(\bar{K}[[t]])\backslash G(\bar{K}((t)))/G(\bar{K}[[t]]) \simeq G(O)\backslash G(F)/G(O)$$

so we omit the subscript of $\mathbf{k}$. For $R \in \mathbf{k}\,\mathrm{Alg}$ and β an isomorphism of G torsors over D_R^*, the relative position is a function locally valued at points, defined as $\mathrm{Inv}_x(\beta) := \mathrm{Inv}(\beta_{(\kappa(x))})$, where $x \in \mathrm{Spec}\, R$ with residue $\kappa(x)$ and $\beta_{\kappa(x)}$ is the base change.
Note that in the language of relative position, a section $(\mathcal{E}, \beta) : \mathrm{Spec}\, R \to \mathrm{Gr}_G$ factors through Gr_μ if and only if $\mathrm{Inv}(\beta) \equiv \mu$. Above argument gives an equivalent definition of Schubert cells (resp. Schubert varieties),

$$\mathrm{Gr}_\mu := \{(\mathcal{E}, \beta) \mid \mathrm{Inv}(\beta) = \mu\}\,; \quad \mathrm{Gr}_{\leqslant\mu} := \{(\mathcal{E}, \beta) \mid \mathrm{Inv}(\beta) \leqslant \mu\}\,.$$

clearly they are subfunctors of Gr_G, and in fact the natural inclusion $\mathrm{Gr}_\mu \hookrightarrow \mathrm{Gr}_{\leqslant\mu}$ is representable by open inclusion. We derive this from

Lemma 1.10. *[Zhu16, Proposition 2.1.4] Let $X := \mathrm{Spec}\, R$, $R \in \mathbf{k}\,\mathrm{Alg}$, fix β, μ as above. Then the locus $\mathrm{Inv}(\beta) \leqslant \mu$ is closed in X, i.e $X_{\leqslant\mu} := \{x \in X \mid \mathrm{Inv}(\beta)(x) \leqslant \mu\} \overset{clo}{\hookrightarrow} X$.*

Theorem 1.11. *Take G a (split) reductive group, fix $T \subseteq G$ maximal torus, for any $\mu \in \mathbb{X}_*(T)^+$,we have:*
*(i) The presheaf Gr_μ is a smooth quasi-projective variety of dimension $(2\rho, \mu)$, we call it **Schubert cell**.*
*(ii) The presheaf $\mathrm{Gr}_{\leqslant\mu}$ is the Zariski closure of Gr_μ, hence a projective variety, we call it **Schubert variety** .*

Proof of Theorem 1.11. Roughly speaking, Gr_μ is the orbit of L^+G left multiplication on the class t^μ in LG. On each $R \in \mathbf{k}\,\mathrm{Alg}$, $L^+G(R) \ni M.(\mathcal{E}, \beta) = (\mathcal{E}, M\beta)$, we can check $\mathrm{Inv}(M\beta) = \mathrm{Inv}(\beta)$. Let $[t^\mu] := t^\mu \cdot L^+G$ be the class of t^μ, for each R, it is the representative of $t^\mu \cdot L^+G(R)$ in $LG(R) = G(R((t)))$.
Define $\underline{\mathrm{Stab}}_{L^+G}(t^\mu)$ to be the sheaf of stabilizer of L^+G-action on $[t^\mu]$, we have

$$\underline{\mathrm{Stab}}_{L^+G}(t^\mu)(R) = \{g \in G(R[[t]]) \mid gt^\mu = t^\mu h, \text{ for some } h \in G(R[[t]])\} = \left(L^+G \cap t^\mu L^+G t^{-\mu}\right)(R)$$

The quotient $[L^+G/\underline{\mathrm{Stab}}_{L^+G}(t^\mu)]$ is an algebraic space, and there is a map

$$[L^+G/\underline{\mathrm{Stab}}_{L^+G}(t^\mu)] \to [LG/L^+G] \simeq \mathrm{Gr}_G$$

sending $g \mapsto gt^\mu$. Étale locally using the identification in the proof of Proposition 1.7, the image locates in Gr_μ. The map is essentially surjective thanks to Cartan decomposition. The tangent space of the homogeneous space Gr_μ is thus $\mathrm{Lie}(L^+G)/\mathrm{Lie}(L^+G \cap t^\mu L^+G t^{-\mu})$, which, using classical theory of affine Lie algebra and loop groups, can be identified with

$$\mathfrak{g}(O)/\mathfrak{g}(O) \cap \mathrm{Ad}_{t^\mu}\, \mathfrak{g}(O) = \bigoplus_{\langle\alpha,\mu\rangle \geqslant 0} \mathfrak{g}_\alpha(O)/\mathrm{t}^{\langle\alpha,\mu\rangle}\mathfrak{g}_\alpha(O)$$

Where $\mathfrak{g} = \bigoplus_\alpha \mathfrak{g}_\alpha$ is Lie algebra of G with root decomposition, and t^μ adjointly acting on $\mathfrak{g}(O) = \mathfrak{g} \otimes \mathbf{k}[[t]]$ by scalar multiplication of $t^{\langle \alpha,\mu \rangle}$ on each graded piece $\mathfrak{g}_\alpha(O)$. Hence we see Gr_μ is smooth of dimension $\langle 2\rho, \mu \rangle$. This proves (i).

For (ii), there are ways to directly prove $\mathrm{Gr}_{\leqslant\mu}$ is closed, for example we can construct Demazure resolution, when $G = \mathrm{GL}_n$, it will be successive (usual) Grassmannian-bundles. Here we introduce another method by Zhu. Consider if $\lambda < \mu$, from the definition of the partial order on coweight lattice, we know there is a positive coroot α such that $\lambda \leqslant \mu - \alpha < \mu$, $\mu - \alpha$ is moreover dominant and $t^{\mu-\alpha} \notin \mathrm{Gr}_\mu$. Want to construct a curve C that is contained in Gr_μ except one point $t^{\mu-\alpha} \in C$.

For any integer m, take $t^{\lambda_m} := \mathrm{diag}(t^m, 1) \in \mathrm{PGL}_2$ and let K_m to be the orbit $\mathrm{Ad}_{t^{\lambda_m}}(L^+\mathrm{SL}_2) \subseteq L\,\mathrm{SL}_2$. Let $L^>\mathrm{SL}_2$ be the kernel of the evaluation map $ev : L^+\mathrm{SL}_2 \to \mathrm{SL}_2$, define $K_m^{(1)} := \mathrm{Ad}_{t^{\lambda_m}}(L^>\mathrm{SL}_2) \subseteq K_m$, we have $K_m/K_m^{(1)} \simeq \mathrm{SL}_2$.

Each coroot α gives an $\mathfrak{sl}_2$ triple in $\mathfrak{g}$, it gives rises to a map $i_\alpha : \mathrm{SL}_2 \to G$, which induces the loop map $Li_\alpha : L\,\mathrm{SL}_2 \to LG$, we take $C_{(\mu,\alpha,m)} := Li_\alpha(K_m) \cdot [t^\mu]$, then $Li_\alpha(K_m^{(1)}) \in \underline{\mathrm{Stab}}_{L^+G}(t^\mu) = L^+G \cap t^\mu L^+ G t^{-\mu}$. Thus $C_{\mu,\alpha}$ is a homogeneous space under the action of $K_m/K_m^{(1)} = \mathrm{SL}_2$, so it is either a point or the projective line $\mathbb{P}^1$. We have

- $C_{\mu,\alpha} = \mathbb{P}^1$.
- $(L^+G \cap Li_\alpha(K_m))([t^\mu]) \simeq \mathbb{A}^1 \subseteq C_{\mu,\alpha}$
- $\sigma_m \in K_m \backslash K_m^{(1)}$, the point $Li_\alpha(\sigma_m) \in \mathbb{P}^1 \backslash \mathbb{A}^1$ can be calculated, it is exactly $t^{\mu-\alpha} \bmod L^+G$.

We proves $\mathrm{Gr}_{\leqslant\mu}$ is the Zariski closure of Gr_μ, hence a projective variety. □

1.4 Beilinson-Drinfeld Grassmannians

In the previous sections, we observe that affine Grassmannians parametrize the information of G torsors (and trivializations) on a formal disc. If we take a smooth projective curve, such a disc could be understood as neighbour of a point on the curve. This fact provides a global vision of affine Grassmannians.

Now let X be a smooth projective curve over $\mathbf{k}$, $x \in |X|$ be a closed point on it. Denote by O_x be the completion of the local ring at x with F_x its fraction field. A choice of coordinate identifies O_x with $\mathbf{k}'[[t]]$, we set $G_x := G \otimes O_x$. Let X_R denote the base change $X \times_{\mathrm{Spec}\,\mathbf{k}} \mathrm{Spec}\, R$ for $R \in \mathbf{k}\,\mathrm{Alg}$, and $X_R^* := (X \backslash x)_R$. Define $\mathrm{Gr}_{G,x}$ to be the functor:

$$R \in \mathbf{k}\,\mathrm{Alg} \mapsto \left\{ (\mathcal{E}, \beta) \mid \mathcal{E} \text{ is a } G\text{-torosr on } X_R; \beta : \mathcal{E}\,|_{X_R^*} \xrightarrow{\simeq} \mathcal{E}^0\,|_{X_R^*} \text{ a trivialization on } X_R^*. \right\}$$

Since there is a natural restriction of G-torsor on X_R to the disc $D_{x,R} = \mathrm{Spec}\, O_x$, we have a morphism of presheaves $\mathrm{Gr}_{G,x}$ to Gr_{G_x} (we mentioned in a remark of Theorem 1.5 that Gr can be defined for smooth affine $\mathbf{k}[[t]]$ group schemes). Notice that trivialization is inherited from restriction. We have the following uniformization theorem à la Beauville-Laszlo:

Theorem 1.12. *The above restriction map* $\mathrm{Gr}_{G,x} \longrightarrow \mathrm{Gr}_{G_x}$ *is an isomorphism.*

Recall the affine Grassmannian Gr_{G_x} can be written as the stack quotient $[LG_x/L^+G_x]$, then the above theorem reduces to show the following identification

$$LG_x(R) \simeq \{(\mathcal{E}, \alpha, \beta) \mid \mathcal{E} \text{ is a } G\text{-torosr on } X_R; \alpha \text{ trivialization of } \mathcal{E}\mid_{D_{x,R}} \beta \text{ trivialization on } X_R^*.\}$$

which boils down to the descent theorem of Beauville-Laszlo.

Proposition 1.13 (Beauville-Laszlo). *Let $\varphi : R \to R'$ be a ring homomorphism, $f \in R$ a nonzerodivisor. Assume $\varphi(f)$ also a nonzerodivisor of R' and that φ induces an isomorphism $R/f \xrightarrow{\simeq} R'/\varphi(f)$. Denote by $\mathrm{Mod}_f(R)$ (resp. $\mathrm{Mod}_{\varphi(f)}(R')$) the abelian category of f-torsion free R-modules (resp. $\varphi(f)$-torsion free R'-modules), then there is an equivalence of categories*

$$\mathbb{F} : \mathrm{Mod}_f(R) \longrightarrow \mathcal{M}$$

here $\mathcal{M}$ denotes the category of triples (M_1, M_2, ϕ), where $M_1 \in \mathrm{Mod}_{R_f}, M_2 \in \mathrm{Mod}_{\varphi(f)}(R')$ and $\phi : M_1 \otimes_{R_f} R'_{\varphi(f)} \xrightarrow{\simeq} (M_2)_{\varphi(f)}$. The functor $\mathbb{F}$ is given by

$$M \mapsto \left(M_f, M \otimes_R R', \phi : M_f \otimes_{R_f} R'_{\varphi(f)} \xrightarrow{\simeq} (M \otimes_R R')_{\varphi(f)}\right)$$

Moreover, under such equivalence, $M \in \mathrm{Mod}_F(R)$ reconstructed from the triple (M_1, M_2, ϕ) is flat if and only if M_1 and M_2 are.

For a proof and generalization of Beauville-Laszlo descent theorem, see [Sta, Tag 0BNI]

Proof of Theorem 1.12. Turning back to the proof of Theorem 1.12, we can now use Proposition 1.13 to glue G-torsors. Reduce to the open affine containing x, we could set $X = \mathrm{Spec}\, A$ with coordinate t and x corresponds to the zero point. For any $R \in \mathbf{k}\,\mathrm{Alg}$, consider the ring homomorphism $\varphi : A \otimes_{\mathbf{k}} R \to R[[t]]$, where on each side t is a nonzerodivisor and quotient by t remains R.
Let $G = \mathrm{Spec}\, B$ be smooth affine group scheme over $\mathrm{Spec}\, A$, consider two modules, $M_1 := B \otimes_A (A[t^-1] \otimes R)$ and $M_2 := B \otimes_A R[[t]]$. We know M_1 is flat over $A[t^-1] \otimes R$ and M_2 is flat over $R[[t]]$. Moreover, $(M_1) \otimes_{A[t^{-1}]\otimes R} R((t)) \simeq B \otimes_A R((t)) = (M_2)_t$. We take a nontrivial isomorphism ϕ between two modules induced by $g \in LG_x(R) = G_x\,(R((t)))$, concretely, it is the sequence

$$\phi : B \otimes_A R((t)) \longrightarrow (B \otimes_A \mathcal{O}_x) \otimes_{\mathcal{O}_x} (B \otimes_A \mathcal{O}_x) \xrightarrow{\mathrm{id}} \otimes g^* B \otimes_A R((t))$$

where $g^* = B \otimes_A \mathcal{O}_x \to R((t))$ given by g. Then by Proposition 1.13, we glue the data (M_1, M_2, ϕ) to a t-regular algebra H flat over $A \otimes R$. Set $\mathcal{E} := \mathrm{Spec}\, H$, it endowed with a G-action and is a G-torsor restricted to $D_{x,R}$ and X_R^*, let α, β be the corresponding canonical trivializations. Also $\mathcal{E}$ itself is a G-torsor over the curve X_R, as one can consider $H \otimes_{A\otimes R} H \longrightarrow (B \otimes R) \otimes_{A\otimes R} H$ is an isomorphism. Conversely, we can recover $g \in G_x(R((t)))$ by $(\beta\mid_{D^*_{x,R}}) \circ (\alpha\mid_{D^*_{x,R}})^{-1}$.
Notice by the identification $\mathrm{Gr}_{G_x} \simeq [LG_x/L^+G_x]$, we get the isomorphism by killing the trivialization α. □

We have seen that on a curve X, the modification of torsors on a fixed point x will be identified with the affine Grassmannian. Further, we can add the information of choice of point into the data, namely we allow the point x on curve to move.
Let X be a smooth projective geometrically connected curve over $\mathbf{k}$, define the **Beilinson-Drinfeld Grassmannian** to be

Definition 1.14. $\mathrm{Gr}_{G,X}$ is the presheaf:

$$R \in \mathbf{k}\,\mathrm{Alg} \mapsto \left\{(x, \mathcal{E}, \beta) \mid x \in X(R), \mathcal{E} \text{ a } G \text{ torsor on } X_R, \beta : \mathcal{E}\mid_{X_R \setminus \Gamma_x} \xrightarrow{\simeq} \mathcal{E}^0 \mid_{X_R \setminus \Gamma_x} \text{ a trivialization.}\right\}$$

where Γ_x denotes the image of $x : \mathrm{Spec}\, R \to X$.

More generally, we allow modifications along several points. Let I be a finite set, then set Gr_{G,X^I} to be the presheaf:

$$R \in \mathbf{k}\,\mathrm{Alg} \mapsto \left\{(x, \mathcal{E}, \beta) \mid x \in X^I(R), \mathcal{E} \text{ a } G \text{ torsor on } X_R, \beta : \mathcal{E}\mid_{X_R \setminus \Gamma_x} \xrightarrow{\simeq} \mathcal{E}^0 \mid_{X_R \setminus \Gamma_x} \text{ a trivialization.}\right\}$$

where for $x = (x_i)_{i \in I} \in X^I(R)$, $\Gamma_x = \bigcup_{i \in I} \Gamma_{x_i}$

Remark. The Beilinson-Drinfeld Grassmannians and their variations are important ingredients in many contexts. For example in the proof of geometric Satake equivalence, one need to define the fusion product over spherical perverse sheaves on affine Grassmannians, then one need to pass to Beilinson-Drinfeld Grassmannians. Also they are used to define Shtukas that derive a Langlands spectral decomposition of cuspidal automorphic forms. We shall see the applications later.

Clearly Gr_{G,X^I} maps to X^I by projecting the first factor, and its fiber at one fixed point is the affine Grassmannian by Beauville-Laszlo. Similar to Theorem 1.5, we have the following

Proposition 1.15. *The Beilinson-Drinfeld Grassmannian $\mathrm{Gr}_{G,X^I} \to X^I$ is representable by a ind-scheme, which is ind-finite type. Moreover, when G is reductive, it is representable by ind-projective scheme.*

Proof. We first consider the case when $G = \mathrm{GL}_n$, in which we mimic the proof of Theorem 1.3 by taking the approximation of Gr_{G,X^I} by $\mathrm{Gr}^{(N)}_{G,X^I}$ $(N \in \mathbb{N})$, so that its R point is given by bounded lattices.

$$\mathrm{Gr}^{(N)}_{G,X^I}(R) = \left\{(x, \mathcal{E}) \mid x \in X^I(R), \mathcal{O}(N\Gamma_x) \subseteq \mathcal{E} \subseteq \mathcal{O}(-N\Gamma_x)\right\}$$

the map $\mathcal{E} \mapsto \mathcal{O}(-N\Gamma_x)/\mathcal{E}$ makes $\mathrm{Gr}^{(N)}_{G,X^I} \to X^I$ a closed subscheme of Quot scheme, which is projective scheme of finite type. And $\mathrm{Gr}_{G,X^I} = \mathrm{colim}_N\, \mathrm{Gr}^{(N)}_{G,X^I}$ is thus ind-finite type and ind-projective.
For general smooth affine group schemes (or reductive groups), we use the same argument as in the proof of Theorem 1.5. $\square$

There are parallel properties shared by global version (BD Grassmannians) and affine Grassmannians.

Proposition 1.16. Gr_{G,X^I} *is isomorphic to the* $fppf$ *stack* $[(LG)_{X^I}/(L^+G)_{X^I}]$.

We have not clarified yet the definition of $(LG)_{X^I}$ and $(L^+G)_{X^I}$. Note now for a point $x \in X_R$, we shall consider the completion of X along Γ_x. To be more precise, let $\hat{\Gamma}_x$ be the formal completion of Γ_x, write A_x topological ring of its functions, i.e $\hat{\Gamma}_x = \mathrm{Spf}\, A_x$. We have a map $\pi : \hat{\Gamma}_x \to X_R$. Denote also $\hat{\Gamma}'_x = \mathrm{Spec}\, A_x$, then there is also $\pi' : \hat{\Gamma}_x \to \hat{\Gamma}'_x$, it is proved in [BD, Proposition 2.12.6] that there exists a unique map $i : \hat{\Gamma}'_x \to X_R$ such that $\pi = i \circ \pi'$. We can thus set $\hat{\Gamma}^\circ_x := \hat{\Gamma}'_x \backslash \Gamma_x$, it an analog of open disc $D^*_{x,R}$.

Definition 1.17. Define the loop group and the arc group over X^I to be the functor

$$(LG)_{X^I} : R \mapsto \left\{(x,\beta) \mid x \in X^I(R), \beta \in G(\hat{\Gamma}^\circ_x)\right\}; \quad (L^+G)_{X^I} : R \mapsto \left\{(x,\beta) \mid x \in X^I(R), \beta \in G(\hat{\Gamma}_x)\right\};$$

Proposition 1.18. *The arc group* $(L^+G)_{X^I}$ *is representable by an affine scheme over* X^I*, the loop group* $(LG)_{X^I}$ *is representable by an ind-scheme over* X^I*. Moreover,* $(LG)_{X^I}(R)$ *can be identified with the quadruple*

$$\left(x, \mathcal{E}, \alpha, \beta \mid x \in X^I(R), \mathcal{E} : G\text{-torsor over } X_R, \alpha : \mathcal{E}\mid_{\hat{\Gamma}'_x} \xrightarrow{\simeq} \mathcal{E}^0 \mid_{\hat{\Gamma}'_x}\ \beta : \mathcal{E}\mid_{X_R \backslash \Gamma_x} \xrightarrow{\simeq} \mathcal{E}^0 \mid_{X_R\backslash\Gamma_x}\right)$$

Proof. To prove $(L^+G)_{X^I} \to X^I$ representable by affine scheme, only need to consider for arbitrary affine scheme $\mathrm{Spec}\, R$, $(L^+G)_{X^I} \times_{X^I} \mathrm{Spec}\, R$ is affine. By definition, it is the limit $\lim_i \mathrm{Res}_{\hat{\Gamma}^{(i)}_x/R}\, G$, where $\hat{\Gamma}^{(i)}_x$ is the i-th infinitesimal truncation of $\hat{\Gamma}_x$, hence is affine. Similarly, we see that $(LG)_{X^I} \times_{X^I} \mathrm{Spec}\, R$ is the inductive limit $\mathrm{colim}_i\, \mathrm{Res}_{\hat{\Gamma}_x[\xi^{-i}]/R}\, G$, where we take ξ the uniformizer. $(LG)_{X^I}$ is thus ind-affine over X^I.
The last statement follows easily from Beauville-Laszlo, see [BD, Theorem 2.12.1]. $\square$

Remark. We see that Proposition 1.16 follows from the previous Proposition and the fact that any G-torsor over $\hat{\Gamma}'_x$ can be $fppf$ locally trivialized.

One may wonder the relationship between Gr_{G,X^I} for different index set I, we have the following intuitive identifications.

Proposition 1.19. *There are canonical isomorphisms:*

$$\Delta : \mathrm{Gr}_{G,X} \simeq \mathrm{Gr}_{G,X^2} \times_{X^2,\Delta} X; \quad c : \mathrm{Gr}_{G,X^2}\mid_{X^2\backslash\Delta_X} \simeq (\mathrm{Gr}_{G,X} \times \mathrm{Gr}_{G,X})\mid_{X^2\backslash\Delta_X}.$$

Moreover, the second isomorphism is $\mathfrak{S}_2$ *equivariant, where* $\mathfrak{S}_2$ *acts obviously by switching the factors.*

Proof. The first isomorphism is followed by definition, the intersection $\mathrm{Gr}_{G,X^2} \times_{X^2} X$ means one still modifies at one point.
For the second one, we have the map $\mathrm{Gr}_{G,X^2}\mid_{X^2\backslash\Delta_X}(R) \longrightarrow (\mathrm{Gr}_{G,X}\times\mathrm{Gr}_{G,X})\mid_{X^2\backslash\Delta_X}(R)$ given by:

$$\left(x_1 \neq x_2, \mathcal{E}, \beta : \mathcal{E}\mid_{X-(\Gamma_{x_1}\cup\Gamma_{x_2})} \to \mathcal{E}^0 \mid_{X-(\Gamma_{x_1}\cup\Gamma_{x_2})}\right) \mapsto (x_1, \mathcal{E}_{x_1}, \beta_1) \times (x_2, \mathcal{E}_{x_2}, \beta_2)$$

where $\mathcal{E}_{x_1}$ is the G-torsor on X_R obtained by gluing $\mathcal{E}\mid_{X_R\backslash\Gamma_{x_2}}$ and $\mathcal{E}^0\mid_{X_R\backslash\Gamma_{x_1}}$ through β, $\beta_1 : \mathcal{E}_{x_1}\mid_{X\backslash\Gamma_{x_1}} \to \mathcal{E}^0\mid_{X\backslash\Gamma_{x_1}}$ is thus the corresponding trivialization. The triple on x_2 is similarly constructed.

Conversely, for triples $(x_1,\mathcal{E}_{x_1},\beta_1),(x_2,\mathcal{E}_{x_2},\beta_2)$, we can construct a G-torsor by gluing $\mathcal{E}_{x_1}|_{X\backslash\Gamma_{x_2}}$ and $\mathcal{E}_{x_2}|_{X\backslash\Gamma_{x_1}}$ through $\beta_2^{-1}\circ\beta_1$ (restrict to $X_R\backslash\Gamma_{(x_1,x_2)}$), $\mathcal{E}$ is thus inherited a trivialization along $X_R\backslash\Gamma_{(x_1,x_2)}$. The two maps are inverse to each other. By construction, we find also permuting x_1 and x_2 interchanges the triples $(x_1,\mathcal{E}_{x_1},\beta_1),(x_2,\mathcal{E}_{x_2},\beta_2)$, thus the second isomorphism is $\mathfrak{S}_2$-equivariant. $\square$

Also we have stratification of Beilinson-Drinfeld Grassmannian by coweights, namely Schubert varieties, inherited from the stratification on affine Grassmannians. The procedure introduced in [Zhu16] is the following: Set $\mathrm{Aut}(D)$ be the ind-group scheme of automorphisms on a disc, $\mathrm{Aut}(D)(R)=\mathrm{Aut}_R(R[[t]])$. Also set $\mathrm{Aut}^+(D)\subseteq\mathrm{Aut}(D)$ of those automorphisms are identity restrict to base. Concretely, $\mathrm{Aut}(D)\simeq\mathrm{Spf}\,\mathbf{k}[[a_0]]\times\mathrm{Spec}\,\mathbf{k}[a_1^{\pm},a_2,\ldots]$ and $\mathrm{Aut}^+(D)\simeq\mathrm{Spec}\,\mathbf{k}[a_1^{\pm},a_2,\ldots]$. We have $\mathrm{Aut}(D)$ acting on affine Grassmannian Gr_G by $g.(\mathcal{E},\beta)\mapsto(g^*\mathcal{E},g^*\beta)$. Let $\hat{X}$ over X to be the presheaf that $\hat{X}(R)=\left\{(x,\alpha)\mid x\in X(R),\alpha:\hat{\Gamma}'_x\xrightarrow{\simeq}D_R\right\}$, it turns out to be a $\mathrm{Aut}(D)$-torsor over X. We can set also

$$\hat{X}^+(R)=\left\{(x,\alpha)\mid x\in X(R),\alpha:\hat{\Gamma}'_x\xrightarrow{\simeq}D_R,\alpha|_{\Gamma_x}:\Gamma_x\simeq\mathrm{Spec}\,R\text{ canonical isomorphism}\right\}$$

it is $\mathrm{Aut}^+(D)$-torsor over X. We have isomorphisms

$$\mathrm{Gr}_{G,X}\simeq\hat{X}\times^{\mathrm{Aut}(D)}\mathrm{Gr}_G\simeq\hat{X}^+\times^{\mathrm{Aut}^+(D)}\mathrm{Gr}_G=:X\tilde{\times}\mathrm{Gr}_G$$

Namely we regard $\mathrm{Gr}_{G,X}$ as a twisted product of X and Gr_G. We set $\mathrm{Gr}_{\leqslant\mu,X}:=X\tilde{\times}\mathrm{Gr}_{\leqslant\mu}$ to be the Schubert variety (as a closed subscheme of $\mathrm{Gr}_{G,X}$). It is nothing but classifies the triple $(x,\mathcal{E},\beta)$ such that $\mathrm{Inv}_x(\beta)\leqslant\mu$.

Proposition 1.20. *Assume that the exponent characteristic of $\mathbf{k}\nmid\pi_1(G_{\mathrm{der}})$, let $\mathrm{Gr}_{\leqslant(\mu,\nu),X^2}$ be the closure of $\mathrm{Gr}_{\leqslant\mu,X}\times\mathrm{Gr}_{\leqslant\nu,X}|_{X^2\backslash\Delta_X}\subseteq\mathrm{Gr}_{G,X^2}|_{X^2\backslash\Delta_X}$ in Gr_{G,X^2}, then restricts to Δ_X, $\mathrm{Gr}_{\leqslant(\mu,\nu),X^2}|_{\Delta_X}\simeq\mathrm{Gr}_{\leqslant\mu+\nu,X}$.*

In fact, we have a generalized version of Proposition 1.19

Proposition 1.21 (Factorization Theorem)**.** *Let I, J be two finite index set, assume there is a surjective map of sets $\phi:J\twoheadrightarrow I$. We have naturally a relative diagonal map $\Delta(\phi):X^I\to X^J$, given by $(x_i)_{i\in I}\mapsto(x_{\phi(j)})_{j\in J}$. Then we have a canonical isomorphism (also denoted by $\Delta(\phi)$):*

$$\Delta(\phi):\mathrm{Gr}_{G,X^I}\xrightarrow{\simeq}\mathrm{Gr}_{G,X^J}\times_{X^J,\Delta(\phi)}X^I$$

It is the generalization of the first isomorphism in Proposition 1.19. Moreover, for $K\overset{\phi}{\twoheadrightarrow}J\overset{\psi}{\twoheadrightarrow}I$, we have $\Delta(\psi\circ\phi)=\Delta(\psi)\circ\Delta(\phi)$.
Also, for $\phi:J\twoheadrightarrow I$, there is a partition $J=\coprod_{i\in I}J_i=\coprod_{i\in I}\phi^{-1}(\{i\})$, let $X^{(\phi)}\subseteq X^J$ be the counterpart of the ϕ relative diagonal: $X^{(\phi)}=\left\{(x_j)_{j\in J}\mid x_j\cap x_{j'}=\emptyset\text{ if }\phi(j)\neq\phi(j')\right\}$ then we have analog of c in Proposition 1.19:

$$c(\phi):\mathrm{Gr}_{G,X^J}\times_{X^J}X^{(\phi)}\xrightarrow{\simeq}\left(\prod_{i\in I}\mathrm{Gr}_{G,X^{J_i}}\right)\times_{X^J}X^{(\phi)}$$

For $K \overset{\phi}{\twoheadrightarrow} J \overset{\psi}{\twoheadrightarrow} I$, we have the operad-like relation:

$$\left(\prod_{i \in I} c(\phi_i)\right) \circ c(\psi \circ \phi) \mid_{X^{(\phi)}} \simeq c(\phi)$$

Note $K = \coprod_{i\in I} \coprod_{j \in J_i} \phi^{-1}(j) = \coprod_{i \in I}(\psi \circ \phi)^{-1}(i) =: \coprod_{i\in I} K_i$, we denote $\phi_i : K_i \twoheadrightarrow J_i$.

This Proposition is called the factorization theorem, which clarifies the relationship among the whole non-filtered set $\left\{\mathrm{Gr}_{G,X^I}\right\}$. The proof of Proposition 1.21 is just similar to that of Proposition 1.19, where we can regard each block J_i as digit, we omit the full proof here. One can also consider to "put together" all index sets to get a 'limit' of Gr_{G,X^I}, this is realized by so-called Ran spaces. Roughly, we consider $\mathrm{Ran}(X) = \mathrm{colim}_I X^I$ to be the prespace as a (non-filtered) colimit under surjective morphisms. Define the Ran Grassmannian to be the presheaf (non-representable) :

$$\mathrm{Gr}_{G,\mathrm{Ran}(X)} = \left\{(x, \mathcal{E}, \beta) \mid x \in \mathrm{Ran}(X)(R), \mathcal{E} \text{ a } G \text{ torsor on } X_R, \beta : \mathcal{E} \mid_{X_R \setminus \Gamma_x} \xrightarrow{\simeq} \mathcal{E}^0 \mid_{X_R\setminus\Gamma_x} \text{ a trivialization.}\right\}$$

With the obvious first-factor projection $\mathrm{pr}_{\mathrm{Ran}} : \mathrm{Gr}_{G,\mathrm{Ran}(X)} \to \mathrm{Ran}(X)$, we identify $\mathrm{Gr}_{G,X^I} = X^I \times_{\mathrm{Ran}(X)} \mathrm{Gr}_{G,\mathrm{Ran}(I)}$.

1.5 Semi-infinite orbits

Let $B = TN$ be a Borel in G and N its unipotent. There is another decomposition of LG:

$$LG = \coprod_{\mu \in \mathbb{X}_*(T)} LN \cdot t^\mu \cdot L^+G$$

It is called Iwasawa decomposition because of the analogy to the Iwasawa decomposition of p-adic group $G(F) = G(\mathcal{O})T(F)N(F)$ (as topological group).

Definition 1.22. We define the **semi-infinite orbit** S_λ in Gr_G to be $LN \cdot t^\lambda$.

Another way to think about this is through the pull-push diagram. We have $T = B/U \twoheadleftarrow B \subseteq G$, which induces the diagram between affine Grassmannians:

$$\mathrm{Gr}_T \xleftarrow{p} \mathrm{Gr}_B \xrightarrow{q} \mathrm{Gr}_G$$

We did not say much on the properties of affine Grassmannians associated with general affine group scheme (in particular, non-reductive, for example B), in above case the set of points in Gr_B and Gr_G are bijective, but they are not isomorphic as ind-scheme. Nevertheless Gr_T is easier to describe, it is just the discrete lattice $\mathbb{X}_*(T)$, followed easily from the Iwasawa decomposition of LT. The semi-infinite orbit S_λ then equals to $q\left(p^{-1}(\lambda)\right)$.

Proposition 1.23. *(i) For each $\lambda \in \mathbb{X}_*(T)$, S_λ is an ind-affine ind-scheme, and in particular, it is of infinite dimension.*

(ii) The closure of S_λ in Gr_G is

$$\overline{S_\lambda} = \coprod_{\mu \leqslant \lambda} S_\mu$$

The following properties are crucial to prove geometric Satake equivalence:

Proposition 1.24. *(i) Let $\lambda, \mu \in \mathbb{X}_*(T)$, with $\mu \in \mathbb{X}_*(T)^+$. Denote by $\check{\Lambda}_G$ the coroot lattice of G, then $S_\lambda \cap \mathrm{Gr}_{\leqslant \mu} \neq \emptyset$ if and only if $\lambda \in \mathrm{Conv}(W \cdot \mu) \cap \left(\mu + \check{\Lambda}_G\right)$, where $\mathrm{Conv}(W \cdot \mu)$ means convex hull of the set $W \cdot \mu$.*
(ii) In the above case, the intersection $S_\lambda \cap \mathrm{Gr}_{\leqslant \mu}$ is pure of dimension $\langle \rho, \lambda + \mu \rangle$.

Remark. Although $S_\lambda \cap \mathrm{Gr}_{\leqslant \mu}$ is of pure dimension, by no mean it is connected!

To be done with MV-cycles and crystal bases.

2 Geometric Satake Equivalence

In this section, without specifying, G always denotes a reductive group over field $\mathbf{k}$. The coefficient of sheaves and complexes could be $\Lambda = \mathbb{Z}/\ell^{\nu}(\nu \geqslant 1)\mathbb{Z}, \mathbb{Z}_\ell, \mathbb{Q}_\ell, \bar{\mathbb{Q}}_\ell$ with ℓ coprime to char $\mathbf{k}$. We state here the arch-goal of this section, which is well-known as the Geometric Satake equivalence:

Theorem 2.1. *There is an equivalence of symmetric monoidal categories*

$$\mathrm{Sat}_G := (\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G), \star) \cong \left(\mathrm{Rep}_{\check{G}}, \otimes\right)$$

where $\check{G}$ is the Langlands dual group of G.

Now we try to let the players who occur in the Theorem 2.1 to show up.

2.1 The Satake category

We see on the LHS of Theorem 2.1, the category is be defined as L^+G-equivariant perverse sheaves on the affine Grassmannian. Let X be a scheme over $\mathbf{k}$, with an action of algebraic group G: $\sigma : G \times X \to X$. For convenience, we say X is a G-space. Denote by $\mathrm{pr}_2 : G \times X \to X$ the natural projection map. A G-equivariant sheaf on X is a pair $(\mathcal{F}, \theta)$, where $\mathcal{F}$ is a sheaf on X and $\theta : \sigma^*\mathcal{F} \simeq \mathrm{pr}_2^*\mathcal{F}$ an isomorphism that satisfies the cocycle condition

$$(\mathrm{id}_G \times \sigma)^*\theta \circ \mathrm{pr}_{23}^*\theta \simeq (m \times id_X)^*\theta; \quad \mathrm{pr}_{23} : G \times G \times X \to G \times X, (g, h, x) \mapsto (h, x), m : G \times G \to G \text{ multiplication}$$

Notice that if G action on X is free, then we have a scheme theoretic good quotient X/G and an equivalence of abelian categories $\mathrm{Sh}_G(X) \cong \mathrm{Sh}(X/G)$. Similarly, we can define the abelian category G-**equivariant perverse sheaves** $\mathrm{Perv}_G(X)$ to have objects $(\mathcal{F}, \theta)$, where $\mathcal{F} \in \mathrm{Perv}(X), \theta : \sigma^*\mathcal{F} \xrightarrow{\simeq} \mathrm{pr}_2^*\mathcal{F}$ that satisfies the cocycle condition as above. The morphism between $(\mathcal{F}, \theta_{\mathcal{F}})$ and $(\mathcal{G}, \theta_{\mathcal{G}})$ is the commutative square

$$\begin{tikzcd}
\sigma^*(\mathcal{F}) \arrow[r, "\theta_F"] \arrow[d] & \mathrm{pr}_2^*(\mathcal{F}) \arrow[d] \\
\sigma^*(\mathcal{G}) \arrow[r, "\theta_G"'] & \mathrm{pr}_2^*(\mathcal{G})
\end{tikzcd}$$

It can be verified that $\mathrm{Perv}_G(X)$ is an abelian category, however such definition is NOT functorial. Indeed, we define the (abelian) category of perverse sheaf to be the heart of t-structure of constructible complexes, the later has six-functor formalism. The equivariant version of derived category is built in [BL94], in which the six functors behave well. We have canonically a forgetful functor $\mathrm{Perv}_G(X) \to \mathrm{Perv}(X)$, so we also expect that there is a forgetful functor between equivariant derived category and the derived category of coherent sheaves that is t-exact.

Definition 2.2. A G resolution of a G-space X is a morphism $f : P \to X$, where P is a free G-space. And say resolution f is n-acyclic ($n > 0$) if $R^0 f_* f^* \simeq \mathrm{id}$ and $R^i f_* f^*$ vanishes for $i = 1, \ldots, n$.

For any resolution $p : P \to X$, let $q : P \to P/G$, define the category $\mathrm{D}^b_G(X, P)$ to be the category with objects $(\mathcal{F}_1 \in \mathrm{D}^b(X), \mathcal{F}_2 \in \mathrm{D}^b(P/G), \gamma : p^*\mathcal{F}_1 \xrightarrow{\simeq} q^*\mathcal{F}_2)$.

Lemma 2.3. *Let X be any G-space.*
(i) For each $n > 0$, there exists an n-acyclic G-resolution.
(ii) Let I be an interval, set $\mathrm{D}^I_G(X, P)$ be the full subcategory of $\mathrm{D}^b_G(X, P)$ with object $(\mathcal{F}_1, \mathcal{F}_2, \gamma)$ so that $\mathcal{F}_1 \in \mathrm{D}^I(X)$. For P, R two G-resolutions of X with $P \to X$ n-acyclic, $n > |I|$. Let $S := P \times_X R$, then there is an equivalence of categories $\mathrm{pr}_2^* : \mathrm{D}^I_G(X, R) \xrightarrow{\simeq} \mathrm{D}^I_G(X, S)$ induced by the projection $S \to R$.

The above lemma allows us to define for each interval I, the G-equivariant I-bounded category $\mathrm{D}^I_G(X)$: we just take any n-acyclic resolution $p : P \to X$ with $n > |I|$ (always exists by (i) of Lemma 2.3), then $\mathrm{D}^I_G(X) := \mathrm{D}^I_G(X, P)$, it is canonical according to (ii) of Lemma 2.3. Finally, we set the bounded G-equivalent derived category of X to be: $\mathrm{D}^b_G(X) := \lim_I \mathrm{D}^I_G(X)$.

Proposition 2.4 ([BL94])**.** *The category $\mathrm{D}^b_G(X)$ is triangulated, with the forgetful functor $\mathrm{D}^b_G(X) \to \mathrm{D}^b(X)$ and truncation functors which define t-structures on $\mathrm{D}^b_G(X)$ Moreover, the heart $\mathrm{D}^{\leqslant 0}_G(X) \cap \mathrm{D}^{\geqslant 0}_G(X)$ is canonically identified with the abelian category of equivariant sheaves $\mathrm{Sh}_G(X)$.*

Note that it is in general NOT true that $\mathrm{D}^b_G(X)$ could be obtained as the derived category of $\mathrm{Sh}_G(X)$.

Theorem 2.5. *[BL94] Let $\mathrm{D}^b_{G,c}(X) \subseteq \mathrm{D}^b_G(X)$ be the full subcategory which forgets to $\mathrm{D}^b_c(X)$ the constructible complexes, also set ${}^p\mathrm{D}^b_{G,c}(X)^{\geqslant a}$, ${}^p\mathrm{D}^b_{G,c}(X)^{\leqslant b}$ to be those forget to ${}^p\mathrm{D}^b_c(X)^{\geqslant a}$, ${}^p\mathrm{D}^b_c(X)^{\leqslant b}$ respectively. Then $\left({}^p\mathrm{D}^b_{G,c}(X)^{\geqslant 0}, {}^p\mathrm{D}^b_{G,c}(X)^{\leqslant 0}\right)$ defines a t-structure with heart equivalent to $\mathrm{Perv}_G(X)$.*

The definition of equivariant perverse sheaves can be extended to certain cases of pro-algebraic group acting on ind-scheme.

Write ind-scheme $X := \mathrm{colim}_i X_i$ co-filtered inductive limit of schemes, with the maps among them are closed embeddings. Since push-forward along closed embedding maps perverse sheaves to perverse sheaves, we can formulate $\mathrm{Perv}(X) := \mathrm{colim}_i \mathrm{Perv}(X_i)$, which is independent of choice of presentation. For algebraic group G-action, we can choose a presentation of ind-scheme X that each X_i is a G-space, then $\mathrm{Perv}_G(X) = \mathrm{colim}_i \mathrm{Perv}_G(X_i)$ forgets to $\mathrm{Perv}(X)$.

Further, if K is a pro-algebraic group, we impose the condition that $|\pi_0(\mathrm{Stab}_K(x))| < \infty, \forall x$ geometric point. And this condition implies that there exists a presentation of X so that the action of K factors through the action of an algebraic quotient K_i (in particular, K_i can be chose to be the quotient of K by a connected normal subgroup). Further, for X_i closed in X_j, can make K_i to be a quotient of K_j. Set $\mathrm{Perv}_K(X_i) := \mathrm{Perv}_{K_i}(X_i)$, it is in fact the colimit along the choice of K_i. For $X_i \hookrightarrow X_j$, we have $\mathrm{Perv}_{K_i}(X_i) = \mathrm{Perv}_{K_j}(X_i) \to \mathrm{Perv}_{K_j}(X_j)$ as before. Consequently, we can define $\mathrm{Perv}_K(X)$ to be the colimit $\mathrm{colim}_i \mathrm{Perv}_K(X_i)$.

Remark. The above construction fits well for arc group action on affine Grassmannian. Only need to verify for each point, the stabilizer in L^+G has finitely many geometric point. In fact, we know the orbits of L^+G-action on Gr_G are given by $\mathrm{Gr}_\mu \simeq L^+G/(L^+G \cap \mathrm{Ad}(t^\mu)L^+G)$. So for any point, whenever it locates in Gr_μ for some μ, the stabilizer of L^+G-action would be $L^+G \cap \mathrm{Ad}(t^\mu)L^+G$, which is connected. So we can formulate $\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G)$

Definition 2.6. We call the well-defined category $\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G)$ the **Satake category** of G, denoted by Sat_G.

Proposition 2.7. *There is an equivalence of category, induced by the forgetful functor* $\mathrm{Fgt} : \mathrm{Sat}_G \longrightarrow \mathrm{Perv}_{\mathscr{S}}(\mathrm{Gr}_G)$*, where* $\mathrm{Perv}_{\mathscr{S}}(\mathrm{Gr}_G) := \lim_{\mu\in\Lambda^+} \mathrm{Perv}(\mathrm{Gr}_{\leqslant\mu})$ *is the category of perverse sheaves on ind-scheme* Gr_G *given by stratification parametrized by coweights.*

Proof. This Proposition is rather hard to prove. See [MV07, Proposition A.1] or [BR18, Proposition 10.8]. □

Theorem 2.8. *The Satake category is semisimple.*

Proof. According to the previous Proposition, we can restrict ourselves to non-equivariant version of perverse sheaves on Gr_G. To wit the theorem, it is enough to show $\mathrm{Ext}^1(\mathbf{IC}_\mu, \mathbf{IC}_\nu) = 0$ for any pair of positive coweights (μ, ν), or equivalently

$$\mathrm{Hom}_{\mathrm{D}^b_{\mathscr{S}}(\mathrm{Gr}_G)}(\mathbf{IC}_\mu, \mathbf{IC}_\nu[1]) = 0$$

We divide the calculations into three case: Firstly, we take $\mu = \nu$, consider the clopen complements in Gr_G:

$$\begin{tikzcd} \mathrm{Gr}_\mu \arrow[r, hook, "j"] \arrow[dr, "j_\mu"'] & \mathrm{Gr}_{\leqslant\mu} \arrow[d] & \mathrm{Gr}_{\leqslant\mu}\setminus\mathrm{Gr}_\mu \arrow[l, hook', "i"'] \arrow[dl, "i_\mu"] \\ & \mathrm{Gr}_G & \end{tikzcd}$$

we get a triangle: $j_!j_\mu^!\mathbf{IC}_\mu \to \mathbf{IC}_\mu \to i_*i^*\mathbf{IC}_\mu \to [1]$. Apply $\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_G)}(\bullet, \mathbf{IC}_\mu[1])$, we have an exact sequence:

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_G)}(i^*\mathbf{IC}_\mu, i^!\mathbf{IC}_\mu[1]) \to \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_G)}(\mathbf{IC}_\mu, \mathbf{IC}_\mu[1]) \to \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_G)}(j_!\bar{\mathbb{Q}}_\ell[\dim(\mathrm{Gr}_\mu)], \mathbf{IC}_\mu[1])$$

Note all complexes are supported on $\mathrm{Gr}_{\leqslant\mu}$, we thus restrict to considering the Hom on $\mathrm{Gr}_{\leqslant\mu}$. Use the Verdier duality and the perversity of $\mathbf{IC}_\mu$, we can deduce that the first term in that exact sequence vanishes. Moreover,

$$\begin{aligned} \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}(j_!\underline{\bar{\mathbb{Q}}_\ell}_{\mathrm{Gr}_\mu}[\dim(\mathrm{Gr}_\mu)], \mathbf{IC}_\mu[1]) &= \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_\mu)}(\underline{\bar{\mathbb{Q}}_\ell}_{\mathrm{Gr}_\mu}[\dim(\mathrm{Gr}_\mu)], j^!\mathbf{IC}_\mu[1]) \\ &\cong \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_\mu)}(\underline{\bar{\mathbb{Q}}_\ell}_{\mathrm{Gr}_\mu}[\dim(\mathrm{Gr}_\mu)], \underline{\bar{\mathbb{Q}}_\ell}_{\mathrm{Gr}_\mu}[1]) \cong H^1(\mathrm{Gr}_\mu, \bar{\mathbb{Q}}_\ell) = 0 \end{aligned}$$

The last equality is due to the fact that Gr_μ is an affine bundle over the partial flag variety G/P_μ, whose odd degree cohomologies vanish. We hence get the vanishing of the middle term, i.e. $\mathrm{Ext}^1(\mathbf{IC}_\mu, \mathbf{IC}_\mu[1]) = 0$.

Next, we consider the case $\mu > \nu$, which implies that $\mathrm{Gr}_\nu \subseteq \overline{\mathrm{Gr}_\mu}$. Since both $\mathbf{IC}_\mu$ and $\mathbf{IC}_\nu$ are supported in $\mathrm{Gr}_{\leqslant\mu}$, we can reduce to compute $\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}(\mathbf{IC}_\mu, \mathbf{IC}_\nu[1])$. Let j be the composition $\mathrm{Gr}_\nu \hookrightarrow \mathrm{Gr}_{\leqslant\nu} \hookrightarrow \mathrm{Gr}_{\leqslant\mu}$ and $\mathcal{F} = \mathrm{Cone}(\mathbf{IC}_\nu \to j_*\underline{\bar{\mathbb{Q}}_\ell}_{\mathrm{Gr}_\nu})$ (Notice it is no longer a clopen triangle). Apply $\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}(\mathbf{IC}_\mu, \bullet)$, we get

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}(\mathbf{IC}_\mu, \mathcal{F}) \to \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}\left(\mathbf{IC}_\mu, \mathbf{IC}_\nu[1]\right) \to \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}\left(\mathbf{IC}_\mu, j_*\bar{\mathbb{Q}}_{\ell_{\mathrm{Gr}_\nu}}[\dim \mathrm{Gr}_\nu + 1]\right)$$

Follow from definition of perverse t-structure, $j_*\underline{\bar{Q}_\ell}_{\mathrm{Gr}_\nu}$ concentrates in non-negative perverse degree. Also, since

$\mathbf{IC}_\nu$ is the intermediate extension of constant sheaf on Gr_ν, take ${}^p\mathcal{H}^0$, we have the injection $\mathbf{IC}_\nu = {}^p\mathcal{H}^0(\mathbf{IC}_\nu) \hookrightarrow {}^p\mathcal{H}^0(j_*\underline{\bar{Q}_\ell}_{\mathrm{Gr}_\nu})$, thus we know $\mathcal{F}$ concentrates in nonnegative perverse degree, with support in $\overline{\mathrm{Gr}_\mu}\backslash \mathrm{Gr}_\mu \supseteq \overline{\mathrm{Gr}_\nu}$, hence $\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}(\mathbf{IC}_\mu, \mathcal{F}) = 0$. Computation of the third term

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\mu})}\left(\mathbf{IC}_\mu, j_*\bar{\mathbb{Q}}_{\ell\mathrm{Gr}_\nu}[\dim \mathrm{Gr}_\nu + 1]\right) = \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\nu})}\left(j^*\mathbf{IC}_\mu, \bar{\mathbb{Q}}_{\ell\mathrm{Gr}_\nu}[\dim \mathrm{Gr}_\nu + 1]\right)$$

uses the following on affine Grassmannians.

Lemma 2.9 (Parity vanishing)**.** *For any $\lambda \in \mathbb{X}_*(T)^+$, the cohomologies $\mathcal{H}^n(\mathbf{IC}_\lambda)$ vanish unless $n \equiv \dim(\mathrm{Gr}_\lambda) \bmod 2$.*

Note that $j^*\mathbf{IC}_\mu \in {}^p\mathrm{D}^{\leqslant -1}(\mathrm{Gr}_{\leqslant\nu}, \bar{\mathbb{Q}}_\ell)$, hence it concentrates in (usual) cohomological degree $\leqslant -\dim \mathrm{Gr}_\nu - 1$. From parity vanishing, we see that $j^*\mathbf{IC}_\mu$ has non-vanishing cohomology degrees in parity with $\dim \mathrm{Gr}_\mu$, note also since $\mu > \nu$, they differ by an element α in coroot lattice, thus the dimension of orbits differ by $\langle 2\rho, \alpha\rangle \equiv 0 \bmod 2$. Hence $j^*\mathbf{IC}_\mu$ concentrates in degree $\leqslant -\dim \mathrm{Gr}_\nu - 2$. It follows $\mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\nu})}\left(j^*\mathbf{IC}_\mu, \bar{\mathbb{Q}}_{\ell\mathrm{Gr}_\nu}[\dim \mathrm{Gr}_\nu + 1]\right) = 0$, as we want. Taking the Verdier duality, we get also the case when $\mu < \nu$.
Finally, we consider the case when $\mu \not\geqslant \nu$ and $\mu \not\leqslant \nu$. Can take λ so that $\mu, \nu \leqslant \lambda$, with inclusions $i_\mu : \overline{\mathrm{Gr}_\mu} \hookrightarrow \overline{\mathrm{Gr}_\lambda}, i_\nu : \overline{\mathrm{Gr}_\nu} \hookrightarrow \overline{\mathrm{Gr}_\lambda}$. Denote by $Z = \overline{\mathrm{Gr}_\mu} \times_{\overline{\mathrm{Gr}_\lambda}} \overline{\mathrm{Gr}_\nu}$, we have proper base change square

$$\begin{tikzcd} Z \arrow[r, hook, "i"] \arrow[d, hook, "j"] & \overline{\mathrm{Gr}_\nu} \arrow[d, hook, "i_\nu"] \\ \overline{\mathrm{Gr}_\mu} \arrow[r, hook, "i_\mu"] & \overline{\mathrm{Gr}_\lambda} \end{tikzcd}$$

Then we have

$$\begin{aligned} \mathrm{Hom}_{\mathrm{D}^b_{\mathscr{S}}(\mathrm{Gr}_G)}(\mathbf{IC}_\mu, \mathbf{IC}_\nu[1]) &= \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\lambda})}(i_\nu^* i_{\mu*}\mathbf{IC}_\mu, \mathbf{IC}_\nu[1]) \\ &= \mathrm{Hom}_{\mathrm{D}^b(\mathrm{Gr}_{\leqslant\lambda})}(i_* j^*\mathbf{IC}_\mu, \mathbf{IC}_\nu[1]) = \mathrm{Hom}_Z(j^*\mathbf{IC}_\mu, i^!\mathbf{IC}_\nu[1]) \end{aligned}$$

Again use the perversity degree argument, we conclude that $\mathrm{Hom}_{\mathrm{D}^b_{\mathscr{S}}(\mathrm{Gr}_G)}(\mathbf{IC}_\mu, \mathbf{IC}_\nu[1]) = 0$. This completes the proof. □

Remark. The parity vanishing condition is a subtle condition, it is known to be valid only when the coefficient ring is of characteristic 0. For general case, one need to consider the so-called parity sheaves (applicable for more general varieties/stacks), see for example [JMW14] for a description.

2.2 ULA sheaves and Fusion product

We have seen that the Satake category Sat_G is semisimple, and moreover it is in fact equipped with a symmetric monoidal structure. Morally, to have a "tensor product" on Sat_G, we need to keep track of information of sheaves on two copies of affine Grassmannians. A good way to see this is passing to Beilinson-Drinfeld Grassmannians, introduced in Section 1.4. To define a global version of Satake category, we introduce the so-called ULA sheaves. To motivate, let $f : X \to S$ be a smooth proper morphism of schemes, then the ℓ-adic cohomology groups $H^i(X_{\bar{s}}, \mathbb{Q}_\ell)$ at each geometric point form a local system on S. If f is no longer smooth, say with an isolated singularity at a "nice" base S, then we can study the monodromy around the cohomology of the fiber, the defect of isomorphism between cohomology at singular stalk and cohomology at generic base is controlled by so-called vanishing cycles. Concretely, let (S, s, η) be a Henselian trait, i.e S is a strictly Henselian DVR, s the geometric point and η its generic point.

Definition 2.10. Let $f : X \to S$ be a morphism of scheme, with (S, s, η) Henselian trait. Let $\bar{\eta}$ be the geometric point over η (take a separable closure of $\kappa(\eta)$). Consider the diagram: Over $\mathrm{D}^+(X_\eta, \Lambda)$ ($\Lambda = \mathbb{Z}/\ell^k\mathbb{Z}$, with ℓ invertible in S), define the **vanishing cycle functor** to be

$$R\Psi_f : \mathrm{D}^+(X, \Lambda) \longrightarrow \mathrm{D}(X_s, \Lambda), \quad \mathcal{F} \mapsto i^* R\bar{j}_* \left(\mathcal{F} \mid_{X_{\bar{\eta}}}\right)$$

where $i : X_s \hookrightarrow X$ and $\bar{j}$ is the composition $X_{\bar{\eta}} \to X_\eta \hookrightarrow X$. Define also the **vanishing cycle functor** to be

$$R\Phi_f : \mathrm{D}^+(X, \Lambda) \longrightarrow \mathrm{D}(X_s, \Lambda), \quad \mathcal{F} \mapsto \mathrm{Cone}\left(i^*\mathcal{F} \mapsto R\Psi_f(\mathcal{F} \mid_{X_\eta})\right)$$

Note that there is a natural Galois group $\mathrm{Gal}(\bar{\eta}/\eta)$ action over $R\Psi_f$. This algebraic definition is an analog to the classical case, in which one takes a continuous map $f : X \to \mathbb{C}^1$ of complex manifolds, with (possibly) critical value at the point $\{0\}$. In this analytic setting, one can consider the Milnor fiber at $x \in X_0$ to be $F_x := \mathbb{B}_\epsilon(x) \cap X_t, t \ll 1, t \neq 0$, and consider the nearby cycle for X to be $R\Psi_f(\mathcal{F}) = i^* R(\tilde{\pi} \circ j)_* j^* \mathcal{F}$, where

$$\begin{tikzcd}
X_0 \arrow[r, hook, "i"] \arrow[d] & X \arrow[d] & X^* \arrow[l, "j"'] \arrow[d] & \widetilde{X^*} \arrow[l, "\tilde{\pi}"'] \arrow[d] \\
\{0\} \arrow[r] & \mathbb{C} & \mathbb{C}\backslash\{0\} \arrow[l] & \mathbb{C} \arrow[l, "\exp"']
\end{tikzcd}$$

and the Galois group $\hat{\mathbb{Z}}$ of punctured disc $\mathbb{C}^*$ acting on $R\Psi_f(\mathcal{F})$. It is known for this analytic case, the stalk of hypercohomology of nearby cycle satisfies $R\Psi_f(\mathcal{F})_x \cong R\Gamma(F_x, \mathcal{F})$. We see that vanishing cycle hence encodes the defect of isomorphism between cohomologies of Milnor fibers. Also it is known that the support of vanishing cycle of constant sheaf is included in the singular locus of X_0.
Back to the algebraic setting, the Henselian DVR S now plays the role of the disc $\mathbb{C}$ and s the critical point, η the punctured disc. We have $R\Psi_f(\mathcal{F})_x = R\Gamma\left((X_{(x)})_{\eta'}, \mathcal{F}\right)$, where $x \in X_s$ lying over s, $X_{(x)}$ the strict henselization of X at x with $(X_{(x)})_{\eta'}$ its generic fiber. We can see $(X_{(x)})_{\eta'}$ here plays the role of the Milnor fiber.

Remark. Using the techniques of vanishing cycles, it is proved in [Del+72] by Grothendieck a monodromy theorem (unipotency of Galois group action on cohomology of generic fiber). Throughout which we can observe arithmetic natures behind these two functors.

Theorem 2.11. *We can restrict the nearby and vanishing cycle functors to constructible complexes (and perverse sheaves).*
(i)[Deligne's finitude theorem] Assume $f : X \to S$ is of finite type, and $\mathcal{F} \in \mathrm{D}^b_c(X, \Lambda)$, then

$$R\Psi_f, R\Phi_f : \mathrm{D}^b_c(X, \Lambda) \to \mathrm{D}^b_c(X_s, \Lambda)$$

(ii) Follow the above assumption, the shifted functors $R\Psi_f[-1], R\Phi_f[-1]$ commute with Verdier duality $\mathbb{D}$. Moreover

$$R\Psi_f[-1], R\Phi_f[-1] : \mathrm{D}^b_c(X,\Lambda) \to \mathrm{D}^b_c(X_s,\Lambda)$$

is t-exact. In particular, they send perverse sheaves to perverse sheaves: $R\Psi_f[-1], R\Phi_f[-1] : \mathrm{Perv}(X) \to \mathrm{Perv}(X_\eta)$.

A definition of nearby/vanishing cycle functors over the base $\mathbb{A}^1$ over some field $\mathbf{k}$ is due to Beilinson, in Beilinson's theory of 'gluing perverse sheaves', we can present some similarity to the classical case. We give a short review of this:
We are in the situation of the following diagram

$$\begin{tikzcd} Z \arrow[r] \arrow[d] & X \arrow[d, "f"] & U \arrow[l] \arrow[d] \\ \{0\} \arrow[r] & \mathbb{A}^1_{\mathbf{k}} & \mathbb{G}_{m,\mathbf{k}} \arrow[l] \end{tikzcd}$$

Now $\pi_1(\mathbb{G}_{m,\mathbf{k}}, 1)$ plays the role of Galois action, it can be fit into short exact sequence

$$1 \to \pi_1^{\mathrm{geom}}(\mathbb{G}_{m,\mathbf{k}}, 1) \to \pi_1(\mathbb{G}_{m,\mathbf{k}}, 1) \to \mathrm{Gal}(\bar{\mathbf{k}}/\mathbf{k}) \to 1$$

where $\pi_1^{\mathrm{geom}}(\mathbb{G}_{m,\mathbf{k}}, 1) = \pi_1(\mathbb{G}_{m,\bar{\mathbf{k}}}, 1)$ is the inertia group. Let p be the exponent characteristic of $\mathbf{k}$, then $\pi_1^{\mathrm{geom}}(\mathbb{G}_{m,\mathbf{k}}, 1)$ has two part: pro-p part P and prime-to-p quotient $\pi_1^{\mathrm{geom},(p)}(\mathbb{G}_{m,\mathbf{k}}, 1)$. In fact $\pi_1(\mathbb{G}_{m,\mathbf{k}}, 1) = \pi_1^{\mathrm{geom}}(\mathbb{G}_{m,\mathbf{k}}, 1) \rtimes \mathrm{Gal}(\bar{\mathbf{k}}/\mathbf{k})$, where the Galois group action on $\pi_1^{\mathrm{geom},(p)}(\mathbb{G}_{m,\mathbf{k}}, 1)$ is given by multiplying cyclotomic character.
Now we 'kill' the wild ramification part and restrict to the tame part of our nearby cycle functor, that is, we can consider

$$R\Psi_f^{\mathrm{P}} : \mathrm{D}^b(U) \to \mathrm{D}^b(Y_{\bar{\mathbf{k}}})$$

There is $\mathrm{Gal}(\bar{\mathbf{k}}/\mathbf{k})$-action on $Y_{\bar{\mathbf{k}}}$, and an $\pi_1(\mathbb{G}_{m,\mathbf{k}}, 1)$-action on $R\Psi_f^{\mathrm{P}}$ compatible with $\mathrm{Gal}(\bar{\mathbf{k}}/\mathbf{k})$-action. Write T generator of prime-to-p part of inertia $\pi_1^{\mathrm{geom}}(\mathbb{G}_{m,\mathbf{k}}, 1)$, then we have exact triangle:

$$R\Psi_f^{\mathrm{P}} \xrightarrow{T-1} R\Psi_f^{\mathrm{P}} \longrightarrow i^* j_* \longrightarrow [1]$$

Moreover, by some linear algebra, we can decompose $R\Psi_f^{\mathrm{P}} = R\Psi_f^{\mathrm{P,u}} \oplus R\Psi_f^{\mathrm{P,nu}}$ into direct sum of unipotent part and invertible part. Restrict ourselves again to the unipotent part, we can take N to be logarithm of T, so that $R\Psi_f^{\mathrm{P,u}}[-1]$ (write $R\Psi_f^{\mathrm{u}}$ in short) sends $\mathrm{Perv}(U)$ to $\mathrm{Perv}(Z)$, and there is exact triangle:

$$R\Psi_f^{\mathrm{u}} \xrightarrow{N} R\Psi_f^{\mathrm{u}} \longrightarrow i^* j_* \longrightarrow [1]$$

Finally, we can glue up $\mathrm{Perv}(U)$ and $\mathrm{Perv}(Z)$ into $\mathrm{Perv}(X)$ by adding extra data of morphisms.

For a general morphism $f : X \to S$ which maps to a general base, the substitute notion for the vanishing of vanishing cycles would be local acyclicity, defined as following: (see also [Del77] or [Sta, Tag 0GJM])

Definition 2.12. Let $f : X \to S$ be a morphism, $\bar{x} \in X$ a geometric point with image $\bar{s} = f(\bar{x})$. Let $t \in S_{(\bar{s})}$ be a geometric point of strict henselization of S at $\bar{s}$. For $\mathcal{F} \in \mathrm{D}^+(X, \Lambda)$, say $\mathcal{F}$ is **locally acyclic relative to** f at x if

$$R\Gamma(X_{(x)}, \mathcal{F}) \longrightarrow R\Gamma(X_{(x)} \times_{S_{(s)}} t, \mathcal{F})$$

is an isomorphism for any $t \in S_{(s)}$ geometric point. Say $\mathcal{F}$ is **universally locally acyclic (ULA) relative to** f (f-ULA in short) if for any base change $S' \to S$, $f' : X' \to S'$ is locally acyclic relative to pullback of $\mathcal{F}$ at all geometric points of S'.

Remark. We can deduce that when the base S is a strict Henselian trait, $\mathcal{F}$ is locally acyclic relative to f if and only if $R\Phi_f(\mathcal{F}) = 0$.

Example 2.13. (i) If f is smooth, then any locally constant sheaf (i.e complex of sheaves with locally constant hypercohomology) is f-ULA. This is mainly deduced from the smooth base change (of étale topology).
In fact, since base change of smooth morphism is smooth, we only need to consider the f-local acyclicity. By doing base change to $X \times_S S_{(\bar{s})} \to S_{(\bar{s})}$, we may assume that $(S, \bar{s})$ fits into an Henselian trait. Then it follows easily as in [Sta, Tag 0GJQ].
(ii) Assuming that S is locally Noetherian and $\mathcal{F}$ is perfect complex of finite type, then $\mathcal{F}$ is id_S-ULA if and only if it is locally constant on S. In fact $t \rightsquigarrow \bar{s}$ is a generalization map, and the condition for $\mathcal{F}$ to be id-ULA says that all specialization maps are isomorphisms, this is followed from [Sta, Tag 0GKC].
(iii) Let $f : X \to S$ and assume $\mathcal{F}$ is f-ULA, then it will satisfy $\mathbb{D}_{X/S}(\mathcal{F}) \otimes^{\mathbb{L}} f^*\mathcal{G} = R\mathcal{H}om(\mathcal{F}, Rf^!\mathcal{G})$, where $\mathbb{D}_{X/S}(\mathcal{F}) := R\mathcal{H}om(\mathcal{F}, Rf^!\Lambda)$ is the relative Verdier duality. We have seen in (i) that if f is smooth, then Λ will be f-ULA, which implies $Rf^!\Lambda \otimes^{\mathbb{L}} f^*\mathcal{G} = Rf^!\mathcal{G}$.

In [LZ22] they introduced a new categorical approach to realize ULA sheaves as dualizable objects in symmetric monoidal 2-category of cohomology correspondence. With which we can formulate the following properties of ULA-sheaves (and generalize the definition of ULA to adic space settings).

Proposition 2.14. *We assume all morphisms to be separated and of finite presentation, then*

(i) Consider the commutative diagram $\begin{array}{ccc} X & \xrightarrow{h} & Y \\ f\big\downarrow & \swarrow_{g} & \\ S & & \end{array}$ *with h proper morphism. Assume $\mathcal{F} \in \mathrm{D}^b(X, \Lambda)$ is f-ULA, then $Rh_*\mathcal{F}$ is g-ULA. In particular, take $g = \mathrm{id}_Y$, we see that $h : X \to Y$ being proper implies $Rh_*\mathcal{F}$ locally constant for any h-ULA sheaf $\mathcal{F}$.*
(ii) Consider the same diagram as in (i), with h being smooth morphism. Assume $\mathcal{G} \in \mathrm{D}^b(Y, \Lambda)$ is g-ULA, then $h^*\mathcal{G}$ is f-ULA.
(iii) Let $\mathcal{F}_i$ be f_i-ULA ($i = 1, 2$), then the exterior product $\mathcal{F} \boxtimes \mathcal{G}$ is (f_1, f_2)-ULA.

Theorem 2.15. *[Rei13, Proposition IV.2.8] Let $f : X \to S$ be a* **k***-morphism, assume $D \subseteq S$ smooth effective Cartier divisor. Set $Z := f^{-1}(D) \overset{i}{\hookrightarrow} X$ closed embedding with complementary $U := X \backslash Z \overset{j}{\hookrightarrow} X$. Assume*

$\mathcal{F} \in \mathrm{D}_c^b(X,\Lambda)$ is f-ULA and $j^*\mathcal{F}$ is perverse, then
(i) There is an isomorphism $i^*\mathcal{F}[-1] \cong i^!\mathcal{F}[1]$ with both complexes $i^*\mathcal{F}[-1], i^!\mathcal{F}[1]$ are perverse. Moreover, $\mathcal{F}$ is perverse and is the intermediate extension $\mathcal{F} \cong j_{!*}(j^*\mathcal{F})$.
(ii) The complex $i^*\mathcal{F}[-1]$ is $f|_Z$-ULA.

Proof. We can first reduce to the case where D is a principal Cartier divisor. Then we have a morphism $\varphi : S \to \mathbb{A}^1$ so that $\varphi^{-1}(0) = D$, let $g = \varphi \circ f$. We can then consider the nearby cycle functor $R\Psi_g[-1] : \mathrm{Perv}(U) \to \mathrm{Perv}(Z)$, and restrict to its unipotent tame part, i,e $R\Psi_g^{\mathrm{P,u}}[-1]$, where P is the pro-p-part of the inertia group mentioned above.
It is known that

$$R\Psi_g^{\mathrm{P,u}}(\mathcal{F}|_U)[-1] \cong \mathrm{colim}_{a\in\mathbb{N}}\, i^* j_{!*}(\mathcal{F}|_U \otimes g^*\mathcal{L}_a)[-1] = i^*\,\mathrm{colim}_a \mathrm{Ker}\left(j_!(\mathcal{F}|_U \otimes g^*\mathcal{L}_a) \to j_*(\mathcal{F}|_U \otimes g^*\mathcal{L}_a)\right)$$

Let $L_a = \Lambda \oplus \Lambda(-1) \oplus \ldots \oplus F(-a), a \in \mathbb{N}$, it admits action of $\pi_1(\mathbb{G}_{m,\mathbf{k}}, 1)$. Concretely, the prime-to-p part acts on L_a by taking exponential of $N = \begin{pmatrix} 0 & 1 & 0 \\ & \ddots & 1 \\ 0 & & 0 \end{pmatrix} : L_a \to L_a(-1)$, and $\mathrm{Gal}(\bar{\mathbf{k}}/\mathbf{k})$ acts by multiplying cyclotomic character. Then $\mathcal{L}_a$ above will be the local system associates with L_a on $\mathbb{G}_m$.
By projection formula, we can write $j_!(\mathcal{F}|_U \otimes g^*\mathcal{L}_a) = \mathcal{F} \otimes j_! g^*\mathcal{L}_a = \mathcal{F} \otimes g^* j_!\mathcal{L}_a$. Also, by ULA property of $\mathcal{F}|_U$ (we will see soon),

$$j_*(\mathcal{F}|_U \otimes g^*\mathcal{L}_a) = j_*\left(\mathbb{D}(\mathbb{D}(\mathcal{F}|_U) \otimes^{\mathbb{L}} \mathbb{D}(g^*\mathcal{L}_a))\right)[2\dim X] = \mathcal{F} \overset{!}{\otimes} j_* g^!\mathcal{L}_a[2\dim X] = \mathcal{F} \overset{!}{\otimes} g^! j_*\mathcal{L}_a[2\dim X]$$

where we use $\mathcal{F}_1 \overset{!}{\otimes} \mathcal{F}_2$ to denote $\mathbb{D}(\mathbb{D}(\mathcal{F}_1) \otimes^{\mathbb{L}} \mathbb{D}(\mathcal{F}_2))$. Then we find for each $a \in \mathbb{N}$, $j_!(\mathcal{F}|_U \otimes g^*\mathcal{L}_a) \to j_*(\mathcal{F}|_U \otimes g^*\mathcal{L}_a)$ is just the complex $\mathcal{F} \otimes^{\mathbb{L}} g^*(j_!\mathcal{L}_a \to j_*\mathcal{L}_a)$. We can compute its cone $C_{X,a}$ is just $\mathcal{F} \otimes^{\mathbb{L}} g^*\,\mathrm{Cone}(j_!\mathcal{L}_a \to j_*\mathcal{L}_a)$. Those $C_{X,a}$ form a sequence of mapping given by the canonical inclusion $L_a \subseteq L_{a+1}$. And in turns those maps are quasi-isomorphisms as perverse cohomologies of $\mathrm{Cone}(j_!\mathcal{L}_a \to j_*\mathcal{L}_a)$ computes nearby cycles of the constant sheaf on $\mathbb{G}_m$. Hence we see $i^*\mathcal{F}[-1] \cong R\Psi_g^{\mathrm{P,u}}(\mathcal{F}|_U)[-1] \cong i^!\mathcal{F}[1]$. The perversity then follows. The ULA property is just followed from the above arguments. $\square$

We present a construction in [LZ22] and [HS23] for ULA-sheaves. In the rest of this section, without specifying, we always assume the coefficient ring to be $\mathbb{Z}/\ell^\nu\mathbb{Z}, \nu \geqslant 1$ or $\mathbb{Q}_\ell$ or $\bar{\mathbb{Q}}_\ell$.
Fix S a base scheme and let $\mathcal{C}_S$ denote the symmetric monoidal 2-category with objects are those $f : X \to S$ separated and of finite presentation. The morphisms are $\mathrm{Hom}_{\mathcal{C}_S}(X,Y) := \mathrm{D}(X \times_S Y, \Lambda)$, with the composition of morphisms given by

$$\mathrm{D}(X \times_S Y, \Lambda) \times \mathrm{D}(Y \times_S Z, \Lambda) \longrightarrow \mathrm{D}(X \times_S Z, \Lambda), \quad (\mathcal{F}, \mathcal{G}) \mapsto \mathcal{F} \star \mathcal{G} := R\pi_{XZ!}(\pi_{XY}^*\mathcal{F} \otimes^{\mathbb{L}} \pi_{YZ}^*\mathcal{G})$$

Here we use the notion π_{XY} to denote the projection map $X \times_S Y \times_S Z \to X \times_S Y$. In the sequel, we also use the subscript to specify the factor of projection if there is no ambiguity. The identity map id_X would be

$R\Delta_{X/S*}\Lambda = R\Delta_{X/S!}\Lambda$.

The 2-category C_S is symmetric monoidal with the monoidal structure defined as $X \boxtimes Y := X \times_S Y$ with the obvious unit S. The associated colax slice category $s\backslash C_S$ of C_S, denoted by C'_S is the symmetric monoidal 2-category with objects to be the pairs $(X, \mathcal{F}), X \in \mathrm{Obj}(C_S), \mathcal{F} \in \mathrm{Hom}_{C_S}(S, X) = \mathrm{D}(X)$. The morphisms in $\mathrm{Hom}_{C'_S}((X,\mathcal{F}),(Y,\mathcal{G}))$ would be in form of pairs $(\mathcal{H}, \phi)$, where $\mathcal{H} \in \mathrm{Hom}_{C_S}(X,Y) = \mathrm{D}(X \times_S Y)$ and $\phi : \mathcal{H} \star \mathcal{F} = R\pi_{Y!}\left(\pi_X^*\mathcal{F} \otimes^{\mathbb{L}} \mathcal{H}\right) \to \mathcal{G}$. The symmetric monoidal structure on C'_S is given by $(X,\mathcal{F}) \boxtimes (Y,\mathcal{G}) := (X \times_S Y, \mathcal{F} \boxtimes \mathcal{G})$, with unit (S, Λ).

Lemma 2.16. *With the monoidal structure on C'_S as above, the inner-hom in C_S is given by*

$$\underline{\mathrm{Hom}}\left((X,\mathcal{F}),(Y,\mathcal{G})\right) = \left(X \times_S Y, R\mathcal{H}om\left(\pi_X^*\mathcal{F}, R\pi_Y^!\mathcal{G}\right)\right)$$

Proof. This follows from the definition, we want for any $(Z, \mathcal{H})$, there is an equivalence

$$\mathrm{Hom}_{C'_S}\left((X,\mathcal{F}) \boxtimes (Y,\mathcal{G}),(Z,\mathcal{H})\right) \cong \mathrm{Hom}_{C'_S}\left((X,\mathcal{F}), \left(Y \times_S Z, R\mathcal{H}om(\pi_Y^*\mathcal{G}, R\pi_Z^!\mathcal{H})\right)\right)$$

Morphisms on the LHS are given by pairs $(\mathcal{E}, \phi)$, where $\mathcal{E} \in \mathrm{D}(X \times_S Y \times_S Z), \phi : (\mathcal{F} \boxtimes \mathcal{G}) \star \mathcal{E} \to \mathcal{H}$.

Consider the following diagram
$$\begin{tikzcd}
X \times_S Y \times_S Z \arrow[r, "\pi_{YZ}"'] \arrow[d, "\pi_X"'] \arrow[dr, "\pi_Y"] \arrow[rr, bend left, "\pi_Z"] & Y \times_S Z \arrow[r, "\pi_Z"'] \arrow[d, "\pi_Y"'] & Z \\
X & Y &
\end{tikzcd}$$
we have:

$$\begin{aligned}
\phi \in \mathrm{Hom}_Z\left((\mathcal{F} \boxtimes \mathcal{G}) \star \mathcal{E}, \mathcal{H}\right) &= \mathrm{Hom}_Z\left(R\pi_{Z!}\left(\pi_X^*\mathcal{F} \otimes^{\mathbb{L}} \pi_Y^*\mathcal{G} \otimes^{\mathbb{L}} \mathcal{E}\right), \mathcal{H}\right) = \mathrm{Hom}_{X \times Y \times Z}\left(\pi_X^*\mathcal{F} \otimes^{\mathbb{L}} \pi_Y^*\mathcal{G} \otimes^{\mathbb{L}} \mathcal{E}, R\pi_Z^!\mathcal{H}\right) \\
&= \mathrm{Hom}_{X \times Y \times Z}\left(\pi_X^*\mathcal{F} \otimes^{\mathbb{L}} \mathcal{E}, R\mathcal{H}om(\pi_Y^*\mathcal{G}, R\pi_Z^!\mathcal{H})\right) = \mathrm{Hom}_{X \times Y \times Z}\left(\pi_X^*\mathcal{F} \otimes^{\mathbb{L}} \mathcal{E}, R\mathcal{H}om(\pi_{YZ}^*\pi_Y^*\mathcal{G}, R\pi_{YZ}^!R\pi_Z^!\mathcal{H})\right) \\
&= \mathrm{Hom}_{X \times Y \times Z}\left(\pi_X^*\mathcal{F} \otimes^{\mathbb{L}} \mathcal{E}, R\pi_{YZ}^!R\mathcal{H}om(\pi_Y^*\mathcal{G}, R\pi_Z^!\mathcal{H})\right) = \mathrm{Hom}_{Y \times Z}\left(R\pi_{YZ!}(\pi_X^*\mathcal{F} \otimes^{\mathbb{L}} \mathcal{E}), R\mathcal{H}om(\pi_Y^*\mathcal{G}, R\pi_Z^!\mathcal{H})\right) \\
&= \mathrm{Hom}_{Y \times Z}(\mathcal{F} \star \mathcal{E}, R\mathcal{H}om(\pi_Y^*\mathcal{G}, R\pi_Z^!\mathcal{H}))
\end{aligned}$$

This completes the comparison of two sides of Hom-set in C'_S above. □

There is a notion of dualizable objects in symmetric monoidal 2-category:

Definition 2.17. Let $(C, \otimes, \mathbf{1})$ be a symmetric monoidal 2-category. An object X of C is called dualizable if there exists an object $X^\vee$, together with maps $\varepsilon : X \otimes X^\vee \to \mathbf{1}$ and $\eta : \mathbf{1} \to X \otimes X^\vee$, called counit and unit maps such that the compositions

$$X \xrightarrow{\eta \otimes \mathrm{id}_X} X \otimes X^\vee \otimes X \xrightarrow{\epsilon \otimes \mathrm{id}_X} X, \quad X^\vee \xrightarrow{\mathrm{id}_{X^\vee} \otimes \eta} X^\vee \otimes X \otimes X^\vee \xrightarrow{\mathrm{id}_{X^\vee} \otimes \epsilon} X^\vee$$

are isomorphic to the respective identities. The object $X^\vee$ is called the dual of X.

Remark. (i) If $X \in C$ is dualizable with dual object $X^\vee$, then we know $- \otimes X^\vee$ is both left and right adjoint to the functor $- \otimes X$. Moreover, we can thus deduce that in such case $\underline{\mathrm{Hom}}(X, Y) = Y \otimes X^\vee$. We see if X is dualizable,

then its dual object is unique under isomorphism, given by $\underline{\mathrm{Hom}}(X,\mathbf{1})$.
(ii) Conversely, if $\underline{\mathrm{Hom}}(X,\mathbf{1})$ and $\underline{\mathrm{Hom}}(X,X)$ exist and $\mathrm{ev}: X\otimes\underline{\mathrm{Hom}}(X,\mathbf{1})\to\underline{\mathrm{Hom}}(X,X)$ is an isomorphism, then $X\in C$ is dualizable. Here ev is obtained from following adjunctions:

$$\mathrm{Hom}(X\otimes\underline{\mathrm{Hom}}(X,\mathbf{1}),\underline{\mathrm{Hom}}(X,X))=\mathrm{Hom}(X\otimes\underline{\mathrm{Hom}}(X,\mathbf{1})\otimes X,X);\mathrm{Hom}(\underline{\mathrm{Hom}}(X,\mathbf{1})\otimes X,\mathbf{1})=\mathrm{Hom}(\underline{\mathrm{Hom}}(X,\mathbf{1}),\underline{\mathrm{Hom}}(X,\mathbf{1}))$$

We take the identity map of $\underline{\mathrm{Hom}}(X,\mathbf{1})$ to get $\mathrm{ev}':\underline{\mathrm{Hom}}(X,\mathbf{1})\otimes X\to\mathbf{1}$ from the second adjunction. And set $\mathrm{ev}:=\mathrm{id}_X\otimes\mathrm{ev}'$.

Definition 2.18. Let $f: X\to S$ be a separated morphism of finite presentation, let $\mathcal{F}\in \mathrm{D}(X,\Lambda)$. Say $\mathcal{F}$ is f-**universally locally acyclic** (f-ULA in short) if the pair $(X,\mathcal{F})\in C'_S$ is dualizable in the sense of Definition 2.17.
We denote by $\mathrm{D}^{\mathrm{ULA}}(X/S,\Lambda)\subseteq \mathrm{D}(X,\Lambda)$ be the full subcategory of f-relative ULA complexes.

Proposition 2.19. *By definition, we get some immediate properties of ULA sheaves.*
(i) From the remark of Definition 2.17, if $(X,\mathcal{F})$ is dualizable, by Lemma 2.16 its dual object would be

$$\underline{\mathrm{Hom}}\left((X,\mathcal{F}),(S,\Lambda)\right)\cong(X,R\mathcal{H}om(\mathcal{F},Rf^!\Lambda))=(X,\mathbb{D}_{X/S}(\mathcal{F}))$$

We see if $\mathcal{F}$ is f-ULA, then its (relative) Verdier dual is also f-ULA in the sense of above definition.
(ii) Before checking the two notions of ULA sheaves coincide, note that in Definition 2.18, the universality is NOT obvious. In fact, for any $s: S'\to S$ base change, we can consider the pullback of 2-categories: $s^*: C_S\to C_{S'}$ by $X\mapsto X\times_S S'$ and the morphisms $\mathrm{Hom}_{C_S}(X,Y)=\mathrm{D}(X\times_S Y)\mapsto \mathrm{Hom}_{C_{S'}}(X_{S'},Y_{S'})=\mathrm{D}((X\times_S Y)_{S'})$. It is easy to check that this pullback functor is symmetric monoidal, so does the induced functor on colax slice categories. And thus $s^*: C'_S\to C'_{S'}$ preserves dualizable objects, this gives the universality we want.
(iii) From the remark after Definition 2.17, we see if $\mathcal{F}$ is f-ULA, then for any $g: Y\to S$ separated and of finite presentation and $\mathcal{G}\in\mathrm{D}(Y)$, we have an isomorphism $R\mathcal{H}om(\pi_X^*\mathcal{F},R\pi_Y^!\mathcal{G})\simeq\mathbb{D}_{X/S}(\mathcal{F})\boxtimes\mathcal{G}$.
(iv) In fact, $\mathcal{F}\in\mathrm{D}(X)$ is f-ULA if and only if $\pi_1^*\mathcal{F}\otimes^{\mathbb{L}}\pi_2^*\mathbb{D}_{X/S}(\mathcal{F})\to R\mathcal{H}om(\pi_1^*\mathcal{F},R\pi_2^!\mathcal{F})$ is an isomorphism, where π_i is the projection of $X\times_S X$ to i-th factor. This returns to property (iii) in Example 2.13.

Proposition 2.20. *Let $\mathcal{F}\in\mathrm{D}(X,\Lambda)$ be f-ULA with respect to $f: X\to S$. Then for $\bar{x}\in X$ geometric point with image $\bar{s}=f(\bar{x})$ and $t\in S_{(\bar{s})}$ generalization of $\bar{s}$, we have*

$$\mathcal{F}_{\bar{x}}:=R\Gamma(X_{(x)},\mathcal{F})\longrightarrow R\Gamma(X_{(x)}\times_{S_{(s)}}t,\mathcal{F})$$

being an isomorphism. In other words, it will imply the local acyclicity in the sense of Definition 2.12.

Proof. Consider the following diagram:

$$\begin{tikzcd} X_{(\bar{x})}\times_{S_{(\bar{s})}} t \arrow[r, "\pi_X"] \arrow[d, "\pi_t"'] & X_{(\bar{x})} \arrow[d, "f"] \\ t \arrow[r, "g"'] & S_{(\bar{s})} \end{tikzcd}$$

By (iii) of Proposition 2.19, take the pair

(t, Λ), we get an isomorphism $\mathcal{F}\mid_{X_{(\bar{x})}\times_{S_{(\bar{s})}} t} = \pi_X^*\mathcal{F}\otimes^{\mathbb{L}}\pi_t^*\Lambda \cong R\mathcal{H}om(\pi_X^*\mathbb{D}(\mathcal{F}), R\pi^!\Lambda)$, apply $R\pi_{X*}$, we get

$$R\pi_{X*}\left(\mathcal{F}\mid_{X_{(\bar{x})}\times_{S_{(\bar{s})}} t}\right) \cong R\mathcal{H}om\left(\mathbb{D}(\mathcal{F}), R\pi_{X*}R\pi_t^!\Lambda\right) \cong R\mathcal{H}om\left(\mathbb{D}(\mathcal{F}), Rf^!Rg_*\Lambda\right)$$

The last equality is the base change. Take the pair $(S_{(\bar{s})}, Rg_*\Lambda)$, we get another isomorphism:

$$\mathcal{F}\otimes^{\mathbb{L}} f^*Rg_*\Lambda \cong R\mathcal{H}om(\mathbb{D}(\mathcal{F}), Rf^!Rg_*\Lambda)$$

Take the stalk at $\bar{x}$, we get $\left(\mathcal{F}\otimes^{\mathbb{L}} f^*Rg_*\Lambda\right)_{\bar{x}} \cong \mathcal{F}_{\bar{x}}\otimes^{\mathbb{L}}(f^*Rg_*\Lambda)_{\bar{x}} \cong \mathcal{F}_{\bar{x}}\otimes^{\mathbb{L}}(Rg_*\Lambda)_{\bar{s}} \cong \mathcal{F}_{\bar{x}}$.
On the other hand, $\left(R\pi_{X*}\left(\mathcal{F}\mid_{X_{(\bar{x})}\times_{S_{(\bar{s})}} t}\right)\right)_{\bar{x}} \cong R\Gamma\left(X_{(\bar{x})}\times_{S_{(\bar{s})}} t, \pi_X^*\mathcal{F}\right)$ by [Sta, Tag 03Q9]. The two sides thus coincide. □

Remark. Subtleties lie in different setting of derived categories. Note [Sta, Tag 03Q9] states for the case $\mathrm{D}^+(X, \Lambda)$ i.e those complexes are bounded below. We need to take the left completion to get the full argument of the above proposition. The universality under base change follows from (ii) of Proposition 2.19.

Theorem 2.21. *[HS23, Theorem 4.4, Theorem 6.8] (i) Assume $f : X \to S$ separated morphism of finite presentation, then for a perfect constructible complex $\mathcal{F}\in \mathrm{D}_{p,c}(X, \Lambda)$, the following two conditions are equivalent:*
(1) $\mathcal{F}$ is f-ULA.
(2) For any $\bar{x}\in X$ geometric point, lying over $\bar{s} := f(\bar{x})$ and for any $t \to S$ generalization of $\bar{s}$, the isomorphism in Proposition 2.20 holds, i.e $\mathcal{F}_{\bar{x}} := R\Gamma(X_{(x)}, \mathcal{F}) \xrightarrow{\simeq} R\Gamma(X_{(x)}\times_{S_{(s)}} t, \mathcal{F})$ is an isomorphism.
(ii) Let $f : X \to S$ as above, assume also S is irreducible, denote $\eta \subseteq S$ its generic point. Let $j : X_\eta \hookrightarrow X$, then $j^* : \mathrm{Perv}^{ULA}(X/S) \to \mathrm{Perv}(X_\eta)$ is an exact and faithful functor.

Proof. Very difficult, to be added in v-topology settings (or maybe not?). □

Finally, we can turn around to the proof of Proposition 2.14

Proof of Proposition 2.14. Thanks to above theorem, we reduce the original conditions for ULA to being dualizable objects.
(i) We want for any pair $(Z, \mathcal{G})$, $\mathbb{D}_{Y/S}(Rh_*\mathcal{F})\boxtimes\mathcal{G} \cong R\mathcal{H}om\left(\pi_Y^*Rh_*\mathcal{F}, R\pi_Z^!\mathcal{G}\right)$. Consider the diagram:

$$\begin{tikzcd} Y & Y\times Z \arrow[l, "\pi_Y"'] \arrow[r, "\pi_Z"] & Z \\ X \arrow[u, "h"] & X\times Z \arrow[l, "\pi_X"] \arrow[u, "h\times \mathrm{id}_Z"'] \arrow[ur, "\pi_Z"'] & \end{tikzcd}$$

Since $\mathcal{F}$ is f-ULA, $\mathbb{D}_{X/S}(\mathcal{F})\boxtimes\mathcal{G} = \pi_X^*\mathbb{D}_{X/S}(\mathcal{F})\otimes^{\mathbb{L}}\pi_Z^*\mathcal{G} \cong R\mathcal{H}om\left(\pi_X^*\mathcal{F}, R\pi_Z^!\mathcal{G}\right)$. Apply the functor $R(h\times\mathrm{id}_Z)_*$

$$\begin{aligned} R(h\times\mathrm{id}_Z)_*(\pi_X^*\mathbb{D}_{X/S}(\mathcal{F})\otimes^{\mathbb{L}}\pi_Z^*\mathcal{G}) &\cong R(h\times\mathrm{id}_Z)_*\pi_X^*\mathbb{D}_{X/S}(\mathcal{F})\otimes^{\mathbb{L}}\pi_Z^*\mathcal{G} \cong \pi_Y^*Rh_*\mathbb{D}_{X/S}(\mathcal{F})\otimes^{\mathbb{L}}\pi_Z^*\mathcal{G} \\ \cong \pi_Y^*\mathbb{D}_{Y/S}(Rh_*\mathcal{F})\otimes^{\mathbb{L}}\pi_Z^*\mathcal{G} &\cong R(h\times\mathrm{id}_Z)_*R\mathcal{H}om_{X\times Z}\left(\pi_X^*\mathcal{F}, R\pi_Z^!\mathcal{G}\right) \cong R\mathcal{H}om_{Y\times Z}\left(\pi_Y^*Rh_*\mathcal{F}, R\pi_Z^!\mathcal{G}\right) \end{aligned}$$

As we want. The second equality and the last one use proper base change.
(ii) Similar to (i), we use the same diagram, and apply $(h \times \mathrm{id}_Z)^*$ to the ULA isomorphism for $\mathcal{G} \in \mathrm{D}(Y)$.
(iii) follows from a more general easy fact that for $(C, \otimes, \mathbf{1})$ symmetric monoidal 2-category, if $\mathcal{F}, \mathcal{G}$ two dualizable objects, then their product $\mathcal{F} \otimes \mathcal{G}$ is dualizable with dual isomorphic to $\mathcal{F}^\vee \otimes \mathcal{G}^\vee$. $\square$

For $X = \mathrm{colim}_i X_i \to S$ an ind-scheme (we require each X_i separated and of finite presentation over the base S), we say $\mathcal{F} \in \mathrm{D}(X) = \mathrm{colim}\, \mathrm{D}(X_i)$ (feel no guilty to take the colimit as derived ∞-categories $\mathcal{D}(X_i)$) is ULA if $\exists \{\mathcal{F}_i \in \mathrm{D}(X_i)\}$ so that $\mathcal{F} = \mathrm{colim}\, \mathcal{F}_i$ and each $\mathcal{F}_i$ is f_i-ULA, where $f_i : X_i \to S$.

Now we focus on applications of ULA sheaves on geometric Satake:
For each finite index set I, define $\mathrm{Sat}_{X^I} := \mathrm{Perv}^{\mathrm{ULA}}_{(L^+G)_{X^I}}(\mathrm{Gr}_{X^I})$ to be the abelian category of equivariant ULA perverse sheaves. Here Gr_{X^I} is the Beilinson-Drinfeld Grassmannian and $(L^+G)_{X^I}$ is also as in Section 1.4. More precisely, we consider the inclusion $\mathrm{D}^{\mathrm{ULA}}(\mathrm{Gr}_{X^I}) \subseteq \mathrm{D}(\mathrm{Gr}_{X^I})$ which is t-exact, where ULA is relative to the projection map $q^I : \mathrm{Gr}_{X^I} \to X^I$. Taking the heart of perverse t-structure, $\mathrm{Perv}^{\mathrm{ULA}}(\mathrm{Gr}_{X^I}) \subseteq \mathrm{Perv}(\mathrm{Gr}_{X^I})$ to be the full subcategory of ULA perverse sheaves. Finally $\mathrm{Sat}_{X^I} \subseteq \mathrm{Perv}^{\mathrm{ULA}}(\mathrm{Gr}_{X^I})$ is taken to be the full subcategory of equivariant ones.

For two finite index sets I, J and a map $\phi : J \to I$, we have seen there is an associated morphism $X^I \to X^J$ by sending $(x_i)_{i \in I} \mapsto (x_{\phi(j)})_{j \in J}$. Moreover, if J surjects to I by ψ, then Proposition 1.21 tells there is a closed embedding

$$\Delta(\phi) : \mathrm{Gr}_{X^I} \cong \mathrm{Gr}_{X^J} \times_{X^J, \Delta(\psi)} X^I \hookrightarrow \mathrm{Gr}_{X^J}$$

Repeatedly use Theorem 2.15, we have that the shifted pull-back $\Delta(\phi)^*[|J| - |I|] : \mathrm{Perv}^{\mathrm{ULA}}(\mathrm{Gr}_{X^J}) \to \mathrm{Perv}^{\mathrm{ULA}}(\mathrm{Gr}_{X^I})$, plus the obvious $(L^+G)_{X^I}$-equivariance, we get $\Delta(\phi)^\bullet := \Delta(\phi)^*[|J| - |I|] : \mathrm{Sat}_{X^J} \to \mathrm{Sat}_{X^I}$.
We introduce a twisted product (convolution) of Beilinson-Drinfeld Grassmannians naturally arising from data of map ψ and of BD Grassmannians. Write $\psi : J \twoheadrightarrow \{1, 2, \ldots, n\}$ and set $J_i := \phi^{-1}(i)$. For $x \in X^J$, write it as product of $x_i \in X^{J_i}$, define:

$$\mathrm{Gr}_{X^J, \phi} := \mathrm{Gr}_{X^{J_1}} \tilde{\times} \mathrm{Gr}_{X^{J_2}} \tilde{\times} \ldots \tilde{\times} \mathrm{Gr}_{X^{J_n}}$$

to be the presheaf:

$$R \in \mathbf{k}\,\mathrm{Alg} \mapsto \left\{ (x, (\mathcal{E}_i, \beta_i)) \mid x \in X^J(R), \mathcal{E}_i : G \text{ torsor on } X_R, \beta : \mathcal{E}_i \mid_{X_R \backslash \Gamma_{x_i}} \xrightarrow{\cong} \mathcal{E}_{i-1} \mid_{X_R \backslash \Gamma_{x_{i-1}}}, \mathcal{E}_0 = \mathcal{E}^0 \right\}$$

In fact, for twisted product $X \tilde{\times} Y$ we always mean the situation where $E \to X$ is a G-torsor, Y is a G-space with the same G and $X \tilde{\times} Y$ is defined to be the stack $[E \times Y / G]$. In the above convolution twisted product, we take the recursive twisted product of the $(L^+G)_{X^{J_n}}$-torsor $\mathbb{E} \to \mathrm{Gr}_{X^{J_1}} \tilde{\times} \ldots \tilde{\times} \mathrm{Gr}_{X^{J_{n-1}}} \times X^{J_n}$ that classifying a point $(x, (\mathcal{E}_i, \beta_i)_{i=1,\ldots,n-1}) \in \mathrm{Gr}_{X^{J_1}} \tilde{\times} \ldots \tilde{\times} \mathrm{Gr}_{X^{J_{n-1}}}$, a point $y \in X^{J_n}$ and a trivialization $\mathcal{E}_{n-1}$ along $\hat{\Gamma}_y$ with $(L^+G)_{X^{J_n}}$-space $\mathrm{Gr}_{X^{J_n}}$.
Write $m_\phi : \mathrm{Gr}_{X^J, \phi} \to \mathrm{Gr}_{X^J}$ the convolution map $(x, (\mathcal{E}_i, \beta_i)) \mapsto (x, \mathcal{E}_n, \beta_1 \circ \ldots \circ \beta_n)$. We can define the exterior convolution product on Sat_{X^J} as following:
We first get the exterior product of $\mathcal{F} \in \mathrm{Perv}_{(L^+G)_{X^{J_1 \times \ldots \times J_n}}}\left(\mathrm{Gr}_{X^{J_1}} \tilde{\times} \ldots \tilde{\times} \mathrm{Gr}_{X^{J_{n-1}}}\right)$ and $\mathcal{A}_n \in \mathrm{Sat}_{X^{J_n}}$ to obtain $\mathcal{F} \tilde{\boxtimes} \mathcal{A}_n \in \mathrm{Perv}_{(L^+G)_{X^J}}(\mathrm{Gr}_{X^J, \phi})$ (again, the equivariance is clear). Thus for $\mathcal{A}_i \in \mathrm{Sat}_{X^{J_i}}$, we have the product

$\mathcal{A}_1 \tilde{\boxtimes} \dots \tilde{\boxtimes} \mathcal{A}_n$. Define the exterior convolution product to be $\mathcal{A}_1 \boxast \dots \boxast \mathcal{A}_n := (m_\phi)_! \left(\mathcal{A}_1 \tilde{\boxtimes} \dots \tilde{\boxtimes} \mathcal{A}_n\right)$.

Proposition 2.22. *Restrict to Satake categories, the above construction gives a well defined functor of exterior convolution product:*

$$\boxast_{i\in\{1,\dots,n\}} : \mathrm{Sat}_{X^{J_1}} \times \dots \times \mathrm{Sat}_{X^{J_n}} \longrightarrow \mathrm{Sat}_{X^J}; \quad (\mathcal{A}_1, \dots, \mathcal{A}_n) \mapsto \mathcal{A}_1 \boxast \dots \boxast \mathcal{A}_n$$

Moreover, for any finite index set J, we can define inner convolution product $\circledast$ on Sat_{X^J} by

$$\circledast_{i\in I} : \prod_I \mathrm{Sat}_{X^J} \longrightarrow \mathrm{Sat}_{X^J}; \quad (\mathcal{A}_1, \dots, \mathcal{A}_n) \mapsto \mathcal{A}_1 \circledast \dots \circledast \mathcal{A}_n := \Delta(\phi)^\bullet \left(\mathcal{A}_1 \boxast \dots \boxast \mathcal{A}_n\right)$$

where ϕ is the diagonal projection $\prod_I J \twoheadrightarrow J$ with $\Delta(\phi)^\bullet$ shifted pull-back introduced before.

Proof. It follows from several consequence we saw before. First, note that $m_\phi : \mathrm{Gr}_{X^J,\phi} \to \mathrm{Gr}_{X^J}$ restricts to an isomorphism when restricting to the open piece $X^{(\phi)}$ as introduced in 1.21, and by 1.21, $\mathrm{Gr}_{X^J} \times_{X^J} X^{(\phi)} \cong \left(\prod_{i\in I} \mathrm{Gr}_{X^{J_i}}\right) \times_{X^J} X^{(\phi)}$. We can deduce that smooth locally the support $\mathrm{Supp}\left(\mathcal{A}_1 \tilde{\boxtimes} \dots \tilde{\boxtimes} \mathcal{A}_n\right) \cong \mathrm{Supp}(\mathcal{A}_1) \times \dots \times \mathrm{Supp}(\mathcal{A}_n)$. Then Proposition 2.14 (iii) tells us that $\mathcal{A}_1 \tilde{\boxtimes} \dots \tilde{\boxtimes} \mathcal{A}_n$ is ULA.
Since m_ϕ is locally closed, and $m_\phi^* m_{\phi !} \left(\mathcal{A}_1 \tilde{\boxtimes} \dots \tilde{\boxtimes} \mathcal{A}_n\right) \cong \mathcal{A}_1 \tilde{\boxtimes} \dots \tilde{\boxtimes} \mathcal{A}_n$ is ULA, combine with Proposition 2.14 (ii) we can conclude that $\mathcal{A}_1 \circledast \dots \circledast \mathcal{A}_n$ is ULA, thus lies in Sat_{X^J} (we have seen it is a equivariant perverse sheaf). $\square$

Remark. It not hard to see that $(\mathrm{Sat}_X, \circledast)$ is a symmetric monoidal category. It will be our replacement for original Sat_G, based on the following fact

Lemma 2.23. *Consider the functor $(\cdot)_X : \mathrm{Sat}_G \to \mathrm{Sat}_X; \mathcal{A} \mapsto \Lambda \tilde{\boxtimes} \mathcal{A}[1] =: \mathcal{A}_X$ it induces an equivalence of categories.*

The inner convolution $\circledast$ will be called the **fusion product**. In fact, we can endow directly Sat_G a monoidal structure, and the above functor $(\cdot)_X$ turns out to be a symmetric monoidal functor.

2.3 Tannakian Reconstruction

3 Langlands Spectral Decomposition over Function Fields

4 On the B_{dR}^{+}-Grassmannians

Now we move to the non-archimedean case and study the affine Grassmannian in the perfectoid geometry setting. Over complex field, recall that by Proposition 1.7, $\mathrm{Gr}_G(C) = LG(\mathbb{C})/L^+G(\mathbb{C}) = G(\mathbb{C}((t)))/G([[t]])$.

An analog of the Laurent series $\mathbb{C}((t))$ for an algebraically closed non-archimedean field $C/\mathbb{Q}_p$ would be the Fontaine's de Rham period field $B_{\mathrm{dR}}(C)$, which is non-canonically isomorphic to $C((\xi))$, where $\xi \in B_{\mathrm{dR}}$ is a uniformizer. The ring of integer of $B_{\mathrm{dR}}(C)$, is denoted by $B_{\mathrm{dR}}^{+}(C)$, which is an ξ-adically complete DVR.

The rough idea of constructing B_{dR}^{+}-Grassmannian is to make a functor $\mathrm{Gr}_G^{B_{\mathrm{dR}}^{+}} : \mathrm{Perf}_{\mathrm{Spd}\, C} \longrightarrow \mathrm{Set}$, so that $\mathrm{Gr}_G^{B_{\mathrm{dR}}^{+}}$ valued on C is $G(B_{\mathrm{dR}}(C))/G(B_{\mathrm{dR}}^{+}(C))$ as we desired. Here Perf is the replacement of Sch, endowed with étale (or v) topology. To begin with, we introduce some notion in perfectoid geometry.

4.1 Perfectoid Rings and Period Rings

Definition 4.1. We say a topological ring R is **Huber** if there exists an open subring $R^\circ \subseteq R$ so that A is adic with finitely generated ideal of definition.

A Huber ring R (with open adic subring R°)is called **Tate** if there exists a topological nilpotent unit $\varpi \in R^\times \cap R^\circ$, which is called a **pseudo-uniformizer**.

A Tate ring R is called **perfectoid Tate** if the subring A as above is ϖ-adically complete, $p \in \varpi^p R^\circ$ and the Frobenius map $R^\circ/\varpi R^\circ \longrightarrow R^\circ/\varpi^p R^\circ, a \mapsto a^p$ is an isomorphism.

Finally, a non-archimedean field C which is also a perfectoid Tate ring will be especially called a **perfectoid field**.

Remark. (i)In fact, a complete non-archimedean field C with a continuous valuation $|\bullet| : C \to \mathbb{R}_{\geqslant 0}$ such that $|p| < 1$ and $\mathcal{O}_C/p\mathcal{O}_C$ has all p-th power roots will be a perfectoid Tate ring.

(ii) If R is a complete Tate ring in characteristic p, i.e $pR = 0$, then R is perfectoid if and only if R is perfect, namely the Frobenius map is an isomorphism.

Example 4.2. (i) $\mathbb{C}_p := \widehat{\overline{\mathbb{Q}_p}}$ the p-adic completion of algebraic closure of $\mathbb{Q}_p$ is a perfectoid field.

(ii) $\mathbb{Q}_p^{\mathrm{cyc}} := \widehat{\cup_{n\geqslant 1}\mathbb{Q}_p[\zeta_{p^n}]}$ the completion of $\mathbb{Q}_p$ joining all p-th power roots of unity, is a perfectoid field.

(iii) $\widehat{\overline{\mathbb{F}_p((t))}}$ is a perfectoid field of characteristic p. $\mathbb{F}_p((t^{1/p^\infty})) := \widehat{\mathbb{F}_p\left(t^{1/p^\infty}\right)} = \widehat{\cup_{n\geqslant 1}\mathbb{F}_p\left(t^{1/p^n}\right)}$ the t-adic completion of $\mathbb{F}_p\left(t^{1/p^\infty}\right)$ is also a perfectoid field of characteristic p.

Definition 4.3. Let R be any ring, we define the **tilt** of R to be the inverse limit

$$R^\flat := \varprojlim_{x\mapsto x^p} R/pR = \lim\left(R/pR \overset{x^p \leftarrow\!\shortmid x}{\longleftarrow} R/pR \overset{x^p \leftarrow\!\shortmid x}{\longleftarrow} R/pR \longleftarrow \dots\right)$$

Proposition 4.4. *[Sch12, Lemma 3.4(i)] If R is ϖ-adically complete and $p \in (\varpi)$, then we have isomorphisms of monoids*

$$\varprojlim_{x\mapsto x^p} R \xrightarrow{\cong} R^\flat = \varprojlim_{x\mapsto x^p} R/pR \xrightarrow{\cong} \varprojlim_{x\mapsto x^p} R/\varpi R$$

induced by $R \twoheadrightarrow R/\varpi R$ which factors through R/pR.

It is not hard to see that $R^\flat$ is perfect and $\operatorname{char} R^\flat = p$, so exhibiting each element f in $R^\flat$ as a sequence of elements in R/pR, $f = (f^{(1)}, f^{(2)}, ...)$, $f^{(i)} = f^p_{(i+1)}$, there is a well-defined addition on $R^\flat$ given by component wise addition, i.e. $f + g = (f^{(1)} + g^{(1)}, f^{(2)} + g^{(2)}, ...)$, $f^{(i)} + g^{(i)} = (f^{(i+1)} + g^{(i+1)})^p$. However this addition law is not suitable for $\varprojlim_{x\mapsto x^p} R$ if R is not of characteristic p.
Indeed, if R is p-torsion free and p-adically complete, $\varprojlim_{x\mapsto x^p} R$ has ring structure by setting that

$$(a+b)^{(n)} := \lim_{k\to+\infty} (a^{(n+k)} + b^{(n+k)})^{p^k}$$

And the isomorphism of monoids above becomes isomorphism of rings.
Let us recall some basic facts about Witt vectors. Fix a prime p, the p-typical Witt vector functor is a functor $W : \mathbf{Rings} \longrightarrow \delta\mathbf{Rings}$ from category of rings to the category of δ-rings, which is right adjoint to the forgetful functor of reverse direction. When p is invertible in R, $W(R)$ is just $R^{\mathbb{N}}$. Par contre, we have:

Proposition 4.5. *If R is perfect ring of characteristic p, then $W(R)$ is p-torsion free and p-adically complete. Moreover $W(R)/pW(R) \cong R$.*

As a corollary of p-adical completeness, if R is perfect in characteristic p, let $[\bullet] : R \to W(R)$ be the Teichmüller lift, then any element x in $W(R)$ can be uniquely written as $x = \sum_{n\geqslant 0} [a_i]p^i, a_i \in R$.

Definition 4.6. In the same setting as Proposition 4.4, we define Fontaine's ring associated with R to be:

$$\mathbb{A}_{\inf}(R) := W(R^\flat)$$

Notice that $\mathbb{A}_{\inf}$ is also a limit of truncated Witt ring $\mathbb{A}_{\inf}(R) = \varprojlim_F W_r(R^\flat)$. By [BMS18, Lemma 3.2 (iv, v)], the canonical morphisms $\varprojlim_F W_r(R) \longrightarrow \varprojlim_F W_r(R^\flat) \longrightarrow \varprojlim_F W_r(R/\varpi R)$ are isomorphisms when R is ϖ-adically complete and ϖ dividing p. As a result, for such R, we can define Fontaine's θ map:

$$\theta : \mathbb{A}_{\inf}(R) \longrightarrow R$$

Remark. (i) As in (ii) of the remark after Definition 4.1, we see that the notion of perfectoid is an analog of perfectness in $\operatorname{char} p$. In general (coming from [BMS18, Definition 3.5]), we say a ring R is a **perfectoid ring** if R is ϖ-adically complete with respect to a non-zero divisor ϖ, $p \in \varpi^p R$, the Frobenius map on $R/\varpi R$ is surjective and the kernel of $\theta : \mathbb{A}_{\inf}(R) \to R$ is principal. Based on that definition, we can morally regard a perfectoid Tate ring as in the form $R\left[\frac{1}{\varpi}\right]$ for some perfectoid ring (R, ϖ).
(ii) Let R be a perfectoid ring and $\xi \in \operatorname{Ker}(\theta)$. We say ξ is *distinguished* if its Witt coordinate $\left(\xi^{(0)}, \xi^{(1)}, \xi^{(2)}\right)$ satisfies $\xi^{(1)}$ is a unit in $R^\flat$, it is equivalent to say ξ generates $\operatorname{Ker}(\theta)$.
(iii) Any ring R in char p is perfectoid if and only if it is perfect. In that case, $p \in \mathbb{A}_{\inf}(R) = W(R)$ is distinguished, $R = R^\flat = W(R)/pW(R)$. So any tilt of perfectoid ring is again perfectoid.
(iv) Assume that R is a complete Tate ring and $R^+ \subseteq R$ subring of integral elements, then if R is perfectoid , R^+ is perfectoid. Conversely, if R^+ is perfectoid and bounded, then R is perfectoid.

Theorem 4.7 (Tilting correspondence)**.** *Assume R is a perfectoid ring, with tilt $R^\flat$, then the tilting functor $(\bullet)^\flat$ induces equivalences of categories:*

$$(\bullet)^\flat : \{\text{perfectoid } R\text{-algebras}\} \xrightarrow{\cong} \left\{\text{perfectoid } R^\flat\text{-algebras}\right\}$$

where the inverse of $(\bullet)^\flat$ is given by $(\bullet)^\sharp := W(\bullet) \otimes_{\mathbb{A}_{\mathrm{inf}}(R),\theta} R$.

The tilting correspondence theorem triggers a question: given a perfectoid ring R of char p, how to characterize those perfectoid rings whose tilt is R?

Definition 4.8. Let R be a perfectoid ring of char p, define its **untilt** to be a pair (A, ι) such that A is perfectoid and $\iota : A^\flat \cong R$. In particular, (R, id) is an untilt of itself.

Example 4.9 (Untilts of perfectoid fields)**.** Now we fix $C^\flat$ a perfectoid field of characteristic p, with continuous non-archimedean valuation $|\bullet|_{C^\flat}$ and ring of integer $O_C^\flat$ characterized by $|x|_{C^\flat} \leqslant 1$. Use $\mathbb{A}_{\mathrm{inf}}$ to denote $W(O_C^\flat)$ for abbreviation. Assume (K, ι) is an untilt of $C^\flat$, then clearly K is a perfectoid field. We denote O_K its ring of integer, which is p-adically complete, and O_K tilts to $O_C^\flat$.
Apply Theorem 4.7 to O_K, we get a map of monoids $\sharp : O_C^\flat \to O_K$ (NOT a ring homomorphism if char $K = 0$!) It induces a ring homomorphism $\sharp : O_C^\flat \to O_K/pO_K$ (an abuse of notation). And there is θ-map:

$$\mathbb{A}_{\mathrm{inf}} \longrightarrow O_K; \quad \sum_{n\geqslant 0}[c_n]p^n \mapsto \sum_{n\geqslant 0} c_n^\sharp p^n$$

That is compatible with $\sharp$ by quotienting pO_K. This θ map is surjective with principal kernel, generated by some distinguished element ξ, thanks to the properties of perfectoid rings. Conversely, if we have $\xi \in \mathbb{A}_{\mathrm{inf}}$ being distinguished, i.e. its Teichmüller expansion is of the form $\sum_{n\geqslant 0}[\xi^{(n)}]p^n$ so that $\left|\xi^{(0)}\right|_{C^\flat} < 1, \left|\xi^{(1)}\right|_{C^\flat} = 1$. Then $\mathbb{A}_{\mathrm{inf}}/(\xi)$ gives an untilt of $C^\flat$. Concretely, we have:

Proposition 4.10. *Assume $C^\flat$ is perfectoid field,* char $C^\flat = p$ *and $\xi \in \mathbb{A}_{\mathrm{inf}} = W(O_C^\flat)$ is a distinguished element. Let O_K denote $\mathbb{A}_{\mathrm{inf}}/\xi\mathbb{A}_{\mathrm{inf}}$, then O_K is a p-torsion free, p-adically complete integral domain and $K :=$ Frac(O_K) is an untilt of $C^\flat$ of* char $= 0$. *Moreover, $\mathbb{A}_{\mathrm{inf}}$ is ξ-adically complete and $\left|(\xi^{(0)})\right|_K = |p|_K = \left|\xi^{(0)}\right|_{C^\flat} < 1$.*

We deduce that for any distinguished element $\xi \in \mathbb{A}_{\mathrm{inf}}$, we can present

$$\mathbb{A}_{\mathrm{inf}} \cong \varprojlim \left(\mathbb{A}_{\mathrm{inf}}/(\xi) = O_K \twoheadleftarrow \mathbb{A}_{\mathrm{inf}}/(\xi^2) \twoheadleftarrow \mathbb{A}_{\mathrm{inf}}/(\xi^3)\dots\right)$$

We can consider another inverse limit by inverting p of components in the above one. Define

$$B_{\mathrm{dR}}^+ = B_{\mathrm{dR}}^+(K) := \varprojlim \left((\mathbb{A}_{\mathrm{inf}}/(\xi))\,[\frac{1}{p}] = O_K[\frac{1}{p}] \cong K \twoheadleftarrow \left(\mathbb{A}_{\mathrm{inf}}/(\xi^2)\right)[\frac{1}{p}] \twoheadleftarrow \left(\mathbb{A}_{\mathrm{inf}}/(\xi^3)\right)[\frac{1}{p}]\dots\right)$$

called **Fontaine's period ring**. We can show that

Proposition 4.11. *Given $C^\flat$ characteristic p perfectoid field and (K, ι) an untilt. $B_{\mathrm{dR}}^+(K)$ does NOT depend on choice of distinguished element $\xi \in$* Ker(θ). *It is a complete discrete valuation ring, any $\xi \in \mathbb{A}_{\mathrm{inf}}$ that* *determines the untilt K would be its uniformizer, which is in particular a non-zero divisor in $B_{\mathrm{dR}}^{+}(K)$. The quotient $B_{\mathrm{dR}}^{+}(K)/(\xi)$ is a field, isomorphic to the untilt K.*

The Proposition 4.10 has its counterpart for $(\mathbb{Q}_p, \mathbb{Z}_p)$ and its finite extension. We say a p-adically complete $\mathbb{Z}_p$-algebra S is an **integral perfectoid ring** if it is of the form $W(R)/(\xi)$, where R is a perfect algebra over $\mathbb{F}_p$ and ξ is a distinguished element in $W(R)$. More generally, assume $F/\mathbb{Q}_p$ a finite extension with ring of integer O_F, uniformizer π and residue field $\mathbb{F}/\mathbb{F}_p$. Define the relative Witt vector functor:

$$W_{O_F}(\bullet) : \{\text{perfect } \mathbb{F}\text{-algebra}\} \longrightarrow \{\pi\text{-complete and} \pi\text{-torsion free} O_F\text{-algebra}\}$$
$$R \mapsto W(R)\hat{\otimes}_{W(\mathbb{F})} O_F =: W_{O_F}(R)$$

Then a p-adically complete O_F-algebra S is said to be integral perfectoid if it is of the form $W_{O_F}(R)/(\xi)$, where $\xi \in W_{O_F}(R)$ is distinguished. Now the definition of perfectoid Tate ring comes legible:

Lemma 4.12. *Assume R is a perfectoid Tate ring with integral element R^+, then R^+ is an integral perfectoid ring. Conversely, if S is an integral perfectoid ring that is ϖ-adically complete and $\varpi^p \mid p$, then $S[\frac{1}{\varpi}]$ is perfectoid Tate, with integral elements being S. This makes precise (i) and (ii) of previous Remark.*

4.2 Perfectoid Spaces and Diamonds

Recall that (A, A^+) is a Huber pair if A is Huber and $A^+ \subseteq A$ is open, integrally closed in A and elements in A^+ are all power-bounded. For the Huber pair (A, A^+), we define its adic spectrum $\mathrm{Spa}(A, A^+)$ to be the equivalent classes of continuous valuations $|\bullet|$ on A such that $|A^+| \leqslant 1$. For any $x \in \mathrm{Spa}(A, A^+)$, we write $|f(x)|$ the corresponding valuation of x on some element $f \in A$. The topology of $\mathrm{Spa}(A, A^+)$ is given by taking the open sets to be of the form:

$$U_{f,g} := \left\{x \in \mathrm{Spa}(A, A^+) \mid |f(x)| \leqslant |g(x)| \neq 0\right\}, \quad f, g \in A$$

Theorem 4.13. *For (A, A^+) Huber pair, $\mathrm{Spa}(A, A^+)$ is a spectral topological space with respect to open sets constructed as above.*

In particular for any ring R, taking its discrete topology, then we can recover $\mathrm{Spec}\, R$ by $\mathrm{Spa}(R, R)$:

$$\mathrm{Spec}\, R \longrightarrow \mathrm{Spa}(R, R) : \mathfrak{p} \mapsto \left(|\bullet|_{\mathfrak{p}} : R \to \mathrm{Frac}(R/p) \to \{0, 1\}\right) \quad \mathrm{Spa}(R, R) \longrightarrow \mathrm{Spec}\, R : |\bullet| \mapsto \mathrm{Ker}(|\bullet|)$$

In general, the geometry of adic spectrum of arbitrary Huber pair is complicated and it is somehow difficult to consider vector bundles on it, since the notion of structure sheaf does NOT exist in all cases. For a Huber pair (A, A^+), we use X to denote $\mathrm{Spa}(A, A^+)$. Define the rational subsets in X to be of the form

$$U\left(\frac{T}{s}\right) := \{x \in X \mid |t(x)| \leqslant |s(x)| \neq 0, \forall t \in T\}, \text{ where } s \in A, T \subseteq A \text{ finite subset so that } TA \subseteq A \text{ open.}$$

It is not hard to check that a rational subset is open in X under adic topology, also finitely many intersection of rational subsets is again rational. More importantly, it is showed by Huber that

Theorem 4.14. *Let $U \subseteq X$ be any rational subset, then there exists a complete Huber pair, denoted by $\left(\mathcal{O}_X(U), \mathcal{O}_X^+(U)\right)$ that (A, A^+) maps to. Taking the adic spectra of Huber pair morphism, we get* $\mathrm{Spa}\left(\mathcal{O}_X(U), \mathcal{O}_X^+(U)\right) \longrightarrow \mathrm{Spa}\,(A, A^+) = X$ *that factors through $U \subseteq X$ and this map is universal among maps to X that factors through U.*

The pairs $(\mathcal{O}_X(U), \mathcal{O}_X^+(U))$ occur in the above Theorem can be lift to presheaves of topological rings by setting

$$\mathcal{O}_X(W) := \varprojlim_{U \subseteq W \text{ rational}} \mathcal{O}_X(U); \quad \mathcal{O}_X^+(W) := \varprojlim_{U \subseteq W \text{ rational}} \mathcal{O}_X^+(U)$$

Definition 4.15. We say a Huber pair (A, A^+) is **sheafy** if the presheaf $\mathcal{O}_X$ associated with $X = \mathrm{Spa}(A, A^+)$ is a sheaf. In that case, $\mathcal{O}_X^+$ is also a sheaf, which can be checked locally on rational subsets.

Definition 4.16. Define the 'category of **adic spaces**' to have objects being the quadruples $\left(X, \mathcal{O}_X, |\bullet(x)|_{x \in X}, \mathcal{O}_X^+\right)$, where X is a topological space, $\mathcal{O}_X$ is a sheaf of topological rings on X, $|\bullet(x)|$ for each $x \in X$ is an equivalent class of $\mathcal{O}_{X,x}$, and $\mathcal{O}_X^+$ is a subsheaf of topological rings on X, so that $|\bullet(x)|$ restrict to $\mathcal{O}_{X,x}^+$ takes value $\leqslant 1, \forall x \in X$.
In the category of adic spaces, a morphism from $\left(X, \mathcal{O}_X, |\bullet(x)|_{x \in X}, \mathcal{O}_X^+\right)$ to $\left(Y, \mathcal{O}_Y, |\bullet(y)|_{y \in Y}, \mathcal{O}_Y^+\right)$ is $f : X \to Y$ continuous map so that locally compatible with valuation maps.
An **adic space** is an object in the above category so that X carries a covering by $\mathrm{Spa}(A_i, A_i^+)$, where (A_i, A_i^+) is sheafy.

Theorem 4.17. *[SW20, Theorem 7.1.1]*
Let (R, R^+) be a perfectoid Huber pair, namely a Huber pair with R being a perfectoid ring, then we have:
(i) (R, R^+) is sheafy, the tilt $(R^\flat, R^{\flat+})$ is also a perfectoid Huber pair, hence is also sheafy. Let $X^\flat := \mathrm{Spa}(R^\flat, R^{\flat+})$.
(ii) Any rational subset $U \subseteq X$ is again perfectoid. Moreover, the tilting map

$$X \to X^\flat; x \mapsto x^\flat := x \circ \sharp$$

is a homeomorphism of topological spaces which is compatible with rational subsets.

Definition 4.18. A **perfectoid space** is an adic space, which can be covered by affinoids $\mathrm{Spa}(R_i, R_i^+)$ so that for each i, (R_i, R_i^+) is a perfectoid Huber pair.
The category of perfectoid spaces is denoted by Perfd, in which morphisms are the same as adic spaces. The subcategory of Perfd contains perfectoid spaces of characteristic p is denoted by Perf.

Remark. (i) The terminology affinoid perfectoid space is reserved for affinoid $\mathrm{Spa}(R, R^+)$ so that (R, R^+) is perfectoid Huber pair. A perfectoid space which is affinoid, however, might NOT be an affinoid perfectoid space.
(ii) By Theorem 4.17, we have the tilting functor $(\bullet)^\flat : \mathrm{Perfd} \longrightarrow \mathrm{Perf}$. For any $X \in \mathrm{Perfd}$, the tilting equivalence establishes an equivalence between categories Perfd_X and $\mathrm{Perf}_{X^\flat}$.

Next we study several Grothendieck topology on perfectoid spaces. Let us first consider maps between them.

Definition 4.19. (i) We call a map of perfectoid Huber pair $(R, R^+) \longrightarrow (S, S^+)$ **finite étale** if $R \to S$ is finite étale and S^+ is integral closure of R^+ in S. A map $f : X \to Y$ of perfectoid spaces is said to be finite étale if for all $\mathrm{Spa}(R, R^+) \to Y$ open affinoid perfectoid space, the pullback $X \times_Y \mathrm{Spa}(R, R^+)$ is perfectoid affinoid of the form $\mathrm{Spa}(S, S^+)$ and that the map $(R, R^+) \to (S, S^+)$ is finite étale as perfectoid Huber pairs.
(ii) A map $f : X \to Y$ of perfectoid spaces is said to be **étale** if it is locally an open immersion of finite étale map, i.e. for any $x \in X$, there exists an open neighbour U of x, so that the restriction $f|_U : U \to V \subseteq Y$ factors as an open immersion $U \hookrightarrow W$ and a finite étale map $W \to V$.
(iii) For $X \in$ Perfd, its étale site is denoted by $X_{\text{ét}}$, it is equivalent to $\left(X^\flat\right)_{ét}$ by tilting.

Definition 4.20. (i) We call a map of perfectoid Huber pairs $(R, R^+) \longrightarrow (S, S^+)$ **affinoid pro-étale** if it is a completed filtered colimit of étale maps. Namely there exists a filtered system of perfectoid Huber pairs (R_i, R_i^+), such that $\mathrm{Spa}(R_i, R_i^+) \to \mathrm{Spa}(R, R^+)$ is étale and $\mathrm{Spa}(S, S^+) \cong \varinjlim \mathrm{Spa}(R_i, R_i^+)$.
(ii) A map $f : X \to Y$ of perfectoid spaces is **pro-étale** if it is locally affinoid proétale as in (i).

For example if $X \in$ Perfd is a perfectoid space, then for any point $x \in X$, the map $x = \mathrm{Spa}\left(\widehat{\kappa(x)}, \widehat{\kappa(x)}^+\right) \to X$ is proétale, as we can present $\widehat{\kappa(x)}$ as $\widehat{O_{X,x}}$, which is the completion of filtered colimit of localizations.

Definition 4.21. (i) Over the category of perfectoid spaces Perfd (resp. Perf), we define the (big) pro-étale site to have the coverings being families of pro-étale maps $\{f_i : X_i \to X\}_{i \in I}$ such that for any qcqs open $U \subseteq X$, there exists a finite subset $I_U \subseteq I$, each $j \in I_U$ a qcqs open $V_j \subseteq X_j$ so that $U = \cup_{j \in I_U} f_j(V_j)$.
(ii) Over the category of perfectoid spaces Perfd, we define the v-site to have the coverings being families of maps $\{f_i : X_i \to X\}_{i \in I}$ such that for any qcqs open $U \subseteq X$, there exists a finite subset $I_U \subseteq I$, each $j \in I_U$ a qcqs open $V_j \subseteq X_j$ so that $U = \cup_{j \in I_U} f_j(V_j)$ (loose the condition of pro-étaleness in (i)).

As is mentioned before, the pro-étale topology on perfectoid spaces are 'fundamental', as the inclusion of a point is pro-étale. In fact, we can morally imagine the pro-étale topology on Perfd as an analog of étale topology on schemes. And the following of diamond would be the substitute for algebraic space in perfectoid geometry.

Definition 4.22. A **diamond** is a pro-étale sheaf in the form $\mathcal{D} = X/R$, where $X \in$ Perf is a perfectoid space of characteristic p, and R is a pro-étale equivalence relation.

Theorem 4.23. *Parallel to the theorem of Gabber that Any algebraic space is a* fpqc-*sheaf, any diamond is a v-sheaf.*

Definition 4.24. For any adic space X over $\mathrm{Spa}(\mathbb{Z}_p, \mathbb{Z}_p)$, we define its *diamondification* to be a functor:

$$X^\diamond : \mathrm{Perf} \longrightarrow \mathrm{Sets}; S \mapsto \left\{(T, f, \iota) \mid T \in \mathrm{Perfd}, f : T \to X \text{ map of adic spaces}, \iota : S \xrightarrow{\cong} T^\flat\right\}.$$

Remark. If $X \in$ Perfd is a perfectoid space, then by tilting equivalence, we can see that $X^\diamond(\bullet)$ is isomorphic to the functor $\mathrm{Hom}_{\mathrm{Perf}}\left(\bullet, X^\flat\right)$, hence a representable presheaf. Since pro-étale (and v)-topology are sub-canonical, we deduce that $X^\diamond$ is just $X^\flat$.

Example 4.25. We consider the diamondification of $\operatorname{Spa}\mathbb{Q}_p := \operatorname{Spa}(\mathbb{Q}_p,\mathbb{Q}_p)$. Let $\operatorname{Spd}\mathbb{Q}_p$ denote $\left(\operatorname{Spa}(\mathrm{Q}_p)\right)^\diamond$, we claim it is isomorphic to $\operatorname{Spa}\left(\mathbb{F}_p((t^{\frac{1}{p^\infty}}))\right)/\underline{\mathbb{Z}_p^\times}$. First, there is a pro-étale epimorphism $\operatorname{Spa}\mathbb{C}_p^\flat \longrightarrow \operatorname{Spd}\mathbb{Q}_p$. Any $X^\sharp/\operatorname{Spa}\mathbb{Q}_p$ untilt of X which is regarded as a map $X \to \operatorname{Spd}\mathbb{Q}_p$, consider any finite extension $K/\mathbb{Q}_p$, $X^\sharp \otimes_{\mathbb{Q}_p} K$ is perfectoid and the map $\left(X^\sharp \otimes_{\mathbb{Q}_p} K\right)^\flat \to X$ is finite étale. We consider the map

$$\left(X^\sharp \hat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p\right)^\flat = \varprojlim_{K/\mathbb{Q}_p \text{ finite}} \left(X^\sharp \otimes_{\mathbb{Q}_p} K\right)^\flat \longrightarrow X$$

is pro-étale, and there is a map $X \to \operatorname{Spa}\mathbb{C}_p^\flat$. Now the equalizer map $\operatorname{Spa}\mathbb{C}_p^\flat \times_{\operatorname{Spd}\mathbb{Q}_p} \operatorname{Spa}\mathbb{C}_p^\flat \rightrightarrows \operatorname{Spa}\mathbb{C}_p^\flat$ can be written as:

$$\begin{aligned}\operatorname{Spa}\mathbb{C}_p \times_{\operatorname{Spa}\mathbb{Q}_p} \operatorname{Spa}\mathbb{C}_p \rightrightarrows \operatorname{Spa}\mathbb{C}_p &= \varprojlim_{K/\mathbb{Q}_p \text{ finite}} \operatorname{Spa}\mathbb{C}_p \times_{\operatorname{Spa}\mathbb{Q}_p} \operatorname{Spa}(K) \rightrightarrows \operatorname{Spa}\mathbb{C}_p \\ &= \operatorname{Spa}\mathbb{C}_p \times \underline{\operatorname{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)} \rightrightarrows \operatorname{Spa}\mathbb{C}_p\end{aligned}$$

This gives a pro-étale equivalence relation and we can deduce:

$$\operatorname{Spd}\mathbb{Q}_p = \operatorname{Spa}\mathbb{C}_p^\flat/\underline{\operatorname{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)} \cong \operatorname{Spa}\left(((\mathbb{Q}_p^{\mathrm{cyc}})^\flat\right)/\underline{\mathbb{Z}_p^\times} \cong mathbb\operatorname{Spd}\mathbb{Q}_p \operatorname{Spa}\left(\mathbb{F}_p((t^{\frac{1}{p^\infty}}))\right)/\underline{\mathbb{Z}_p^\times}.$$

So that $\underline{\mathbb{Z}_p^\times}$ acts on $\mathbb{F}_p((t^{\frac{1}{p^\infty}}))$ by $\gamma(t) := (1+t)^\gamma - 1$.

By the above example 4.25, we observe that the diamondification of $\operatorname{Spa}\mathbb{Q}_p$ is indeed a diamond, and the tilting equivalence tells us it serves as the terminal object in the equivalence of categories:

$$\mathrm{Perfd}/\mathbb{Q}_p \xrightarrow{\cong} \mathrm{Perf}/\operatorname{Spd}\mathbb{Q}_p$$

Note that $\operatorname{Spd}(\mathbb{Z}_p) := \left(\operatorname{Spd}(\mathbb{Z}_p)\right)^\diamond$ is however not a diamond, but a v-sheaf. In general, what we can say about diamondification is:

Theorem 4.26. *For any analytic adic space X over $\operatorname{Spa}(\mathbb{Z}_p,\mathbb{Z}_p)$, $X^\diamond$ is a diamond. For general adic space X, $X^\diamond$ is a v-sheaf.*

We then take a look at some examples of adic spaces.

Example 4.27 (The adic open unit disc over $\mathbb{Z}_p$)**.** Consider $\mathbb{Z}_p[[T]]$ ring of formal series over $\mathbb{Z}_p$, it is complete under (p,T)-adic topology. Let $X := \operatorname{Spa}(\mathbb{Z}_p[[T]])$. In X, there is a unique point $x_{\mathbb{F}_p}$, whose corresponding valuation has open kernel. In fact, $|\bullet|_{x_{\mathbb{F}_p}}$ is given by the composition map $\mathbb{Z}_p[[T]] \to \mathbb{F}_p \to \{0,1\}$ so that the second map sends units in $\mathbb{F}_p$ to 1. $x_{\mathbb{F}_p}$ is called a *non-analytic point*, it corresponding to the non-generic point in $\operatorname{Spec}\mathbb{Z}_p$.
Let $\mathcal{Y} := X \backslash \left\{x_{\mathbb{F}_p}\right\}$, then all points in $\mathcal{Y}$ are *analytic*, namely their kernel are not open in $\mathbb{Z}_p[[T]]$. For a valuation

$x \in \mathcal{Y}$, the value of $|T(x)|$ and $|p(x)|$ can not be zero simultaneously. Define a map $\kappa : \mathcal{Y} \to [0, \infty]$ by:

$$\kappa(x) := \frac{\log |T(\tilde{x})|}{\log |p(\tilde{x})|}, \tilde{x} \text{ maximal generalization of } x$$

We see that $\kappa(x) = 0$ means $|p(\tilde{x}))| = 0$, equivalent to $|p(x))| = 0$. So the valuation $|\bullet|_x$ factors through mod p quotient, i.e. $\mathbb{F}_p[[T]]$. In the same reason, $\kappa(x) = \infty$ implies that $|\bullet|_x$ factors through $\mathbb{Z}_p$.

Example 4.28 (Relative Fargues-Fontaine curve)**.** Now we take $S \in \mathrm{Perf}$ a perfectoid adic space of characteristic p. We first define the product $S \dot{\times} \operatorname{Spa} \mathbb{Z}_p$. Heuristically, if S is adic spectrum of some perfect $\mathbb{F}_p$-algebra, notice since $\mathbb{Z}_p = W(\mathbb{F}_p)$, so we want to make the product similar to the adic spectrum of $R \otimes \mathbb{Z}_p$, which can be thought as $W(R)$. Concretely, if $S = \operatorname{Spa}(R, R^+) \in \mathrm{Perf}$, choose a pesudo-uniformizer $\varpi \in R^+$, we define the dot product $S \dot{\times} \operatorname{Spa} \mathbb{Z}_p$ to be the locus $\operatorname{Spa}(W(R^+)) \setminus \{[\varpi] = 0\}$.
Inside of $\operatorname{Spa}(W(R^+))$, consider the rational subset $U\left(\frac{p}{[\varpi]^{1/p^n}}\right)$. It is realized as an affinoid (but NOT perfectoid) adic space $\operatorname{Spa}(R_n, R_n^+)$, where $R_n^+ = \left(W(R^+)\left[\frac{p}{[\varpi]^{1/p^n}}\right]\right)^{\wedge}$ ($[\varpi]$-adic completion) and $R_n = R_n^+\left[\frac{1}{[\varpi]}\right]$.
We can show after adding all p-power roots of p, R_n is perfectoid, namely the tensor product $R_n \hat{\otimes}_{\mathbb{Z}_p} \mathbb{Z}_p[p^{1/p^\infty}]$ is a perfectoid ring, with pseudo-uniformizer $\left[\varpi^{1/p^{n+1}}\right]$. So by definition, R_n is sous-perfectoid and (R_n, R_n^+) is sheafy. $\operatorname{Spa}(R_n, R_n^+)_{n \geqslant 1}$ jointly covers $\operatorname{Spa}(W(R^+)) \setminus \{[\varpi] = 0\}$.

5 Geometrization of LLC

References

[AHH19] Jarod Alper, Daniel Halpern-Leistner, and Jochen Heinloth. *Cartan-Iwahori-Matsumoto Decompositions for Reductive Groups*. Feb. 2019. doi: 10.48550/arXiv.1903.00128. arXiv: 1903.00128 [math]. (Visited on 01/28/2023).

[BR18] Pierre Baumann and Simon Riche. *Notes on the Geometric Satake Equivalence*. Mar. 2018. doi: 10.48550/arXiv.1703.07288. arXiv: 1703.07288 [math]. (Visited on 05/19/2023).

[BD] Alexander Beilinson and Vladimier Drinfeld. *Quantization of Hitchin's Integrable System and Hecke Eigensheaves*. http://math.uchicago.edu/~drinfeld/langlands/QuantizationHitchin.pdf. (Visited on 05/06/2023).

[BL94] Joseph Bernstein and Valery Lunts. *Equivariant Sheaves and Functors*. Vol. 1578. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 1994. isbn: 978-3-540-58071-3 978-3-540-48430-1. doi: 10.1007/BFb0073549. (Visited on 05/10/2023).

[BMS18] Bhargav Bhatt, Matthew Morrow, and Peter Scholze. "Integral \$p\$-Adic Hodge Theory". In: *Publications mathématiques de l'IHÉS* 128.1 (Nov. 2018), pp. 219–397. issn: 1618-1913. doi: 10.1007/s10240-019-00102-z. (Visited on 01/20/2024).

[BS17] Bhargav Bhatt and Peter Scholze. *Projectivity of the Witt Vector Affine Grassmannian*. Feb. 2017. doi: 10.48550/arXiv.1507.06490. arXiv: 1507.06490 [math]. (Visited on 12/31/2022).

[Del+72] P. Deligne et al. *Groupes de Monodromie En Géométrie Algébrique*. Vol. 288. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 1972. isbn: 978-3-540-05987-5 978-3-540-37984-3. doi: 10.1007/BFb0068688. (Visited on 05/23/2023).

[Del77] Pierre Deligne. *Cohomologie Etale: Séminaire de Géométrie Algébrique du Bois-Marie SGA 4 1/2*. Ed. by A. Dold and B. Eckmann. Vol. 569. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 1977. isbn: 978-3-540-08066-4 978-3-540-37507-4. doi: 10.1007/BFb0091516. (Visited on 05/23/2023).

[Fal03] Gerd Faltings. "Algebraic Loop Groups and Moduli Spaces of Bundles". In: *Journal of the European Mathematical Society* 5.1 (Mar. 2003), pp. 41–68. issn: 1435-9855. doi: 10.1007/s10097-002-0045-x. (Visited on 01/02/2023).

[HS23] David Hansen and Peter Scholze. *Relative Perversity*. May 2023. arXiv: 2109.06766 [math]. (Visited on 05/30/2023).

[JMW14] Daniel Juteau, Carl Mautner, and Geordie Williamson. "Parity Sheaves". In: *Journal of the American Mathematical Society* 27.4 (Oct. 2014), pp. 1169–1212. issn: 0894-0347, 1088-6834. doi: 10.1090/S0894-0347-2014-00804-3. (Visited on 05/19/2023).

[KV04] Mikhail Kapranov and Eric Vasserot. "Vertex Algebras and the Formal Loop Space". In: *Publications Mathématiques de l'IHÉS* 100 (2004), pp. 209–269. issn: 1618-1913. doi: 10.1007/s10240-004-0023-9. (Visited on 01/19/2023).

[LZ22] Qing Lu and Weizhe Zheng. “Categorical Traces and a Relative Lefschetz–Verdier Formula”. In: *Forum of Mathematics, Sigma* 10 (2022), e10. ISSN: 2050-5094. DOI: 10.1017/fms.2022.2. (Visited on 05/24/2023).

[MV07] Ivan Mirković and Kari Vilonen. “Geometric Langlands Duality and Representations of Algebraic Groups over Commutative Rings”. In: *Annals of Mathematics* 166.1 (July 2007), pp. 95–143. ISSN: 0003-486X. DOI: 10.4007/annals.2007.166.95. (Visited on 05/19/2023).

[PR08] G. Pappas and M. Rapoport. *Twisted Loop Groups and Their Affine Flag Varieties*. Apr. 2008. DOI: 10.48550/arXiv.math/0607130. arXiv: math/0607130. (Visited on 01/19/2023).

[Rei13] Ryan Cohen Reich. *Twisted Geometric Satake Equivalence via Gerbes on the Factorizable Grassmannian*. June 2013. DOI: 10.48550/arXiv.1012.5782. arXiv: 1012.5782 [math]. (Visited on 06/10/2023).

[Sch12] Peter Scholze. “Perfectoid Spaces”. In: *Publications Mathématiques de l'IHÉS* 116 (2012), pp. 245–313. ISSN: 1618-1913. DOI: 10.1007/s10240-012-0042-x. (Visited on 01/20/2024).

[SW20] Peter Scholze and Jared Weinstein. *Berkeley Lectures on P-Adic Geometry: (AMS-207)*. Princeton University Press, 2020. ISBN: 978-0-691-20208-2. DOI: 10.2307/j.ctvs32rc9. JSTOR: j.ctvs32rc9. (Visited on 01/16/2023).

[Sta] Stacks project authors. *The Stacks Project*. https://stacks.math.columbia.edu/. (Visited on 01/03/2023).

[Zhu16] Xinwen Zhu. *An Introduction to Affine Grassmannians and the Geometric Satake Equivalence*. Apr. 2016. DOI: 10.48550/arXiv.1603.05593. arXiv: 1603.05593 [math]. (Visited on 12/31/2022).